

A Theory of Stable Market Segmentations*

Nima Haghpanah[†] Ron Siegel[‡]

This draft: August 16, 2025
First draft: February 10, 2022

Abstract

We study markets in which consumers interact with a monopolistic seller via data-collecting intermediaries that serve consumers' best interests. The interaction leads to a segmentation of the consumers into groups, each represented by an intermediary, and the seller sets an optimal price for each group. We focus on the strategic tension that arises when high-value consumers want to group with low-value consumers in order to obtain a low price. We introduce three notions to capture segmentations that resolve this tension. These notions are (separately) motivated by the following ideas: (1) breaking apart an existing group and forming a new one involve an administrative friction, (2) a consumer's consent is required to aggregate his data, and (3) each intermediary acts on behalf of his group's best interests. Our main result shows that these notions are equivalent and non-empty, and are characterized by efficiency and saturation. A segmentation is saturated if shifting consumers from a group with a higher price to a group with a lower price leads the seller to optimally increase the lower price. We show that stable segmentations that maximizes average consumer surplus (across all segmentations) always exist and discuss policy implications.

Keywords: Market segmentation, Price discrimination, Pooling, Group formation, Stability.

*We thank Rohan Dutta, Matt Mitchell, Debraj Ray, Daniel Rappoport, Doron Ravid, Joe Root, Jakub Steiner, Roland Strausz, Rakesh Vohra, Asher Wolinsky, and participants at various seminars and conferences for comments.

[†]Yale School of Management, New Haven, CT 06520. email: nima.haghpanah@yale.edu.

[‡]Department of Economics, the Pennsylvania State University, University Park, PA 16802. e-mail: rus41@psu.edu.

1 Introduction

A salient feature of the modern economy is firms’ ability to collect detailed consumer data from a large number of consumers. Such data can be used by sellers to effectuate wide-ranging and systematic price discrimination, a possibility with which regulators worldwide are grappling. One remedy discussed for over a decade is giving consumers more control over their data. For example, a 2012 report by the Federal Trade Commission states that “As companies develop new means for collecting data about individuals [...], consumers should have more tools to control data collection, not fewer.”¹

A recent development that may help in this direction is data-collecting intermediaries that pool consumers’ data and facilitate their interactions with sellers. Mediators of individual data, which were recently introduced by Lanier and Weyl (2018), and data cooperatives, such as MiData, Salus Coop, and Swash, serve such a role by collecting and aggregating consumer data while protecting consumers’ rights over their data.² Data cooperatives have a fiduciary duty to act in the best interest of their members, and their primary goal is to benefit their members. This suggests the intriguing prospect of consumers that interact with sellers via data intermediaries that serve consumers’ best interests. Such practices are in their infancy, and raise the question of whether such intermediaries are likely to arise and serve an effective role, and, if so, what market outcomes will result as a consequence.

We investigate these questions by considering a market in which consumers interact with a monopolistic seller via intermediaries. Each consumer has one of a finite number of possible values for the seller’s product and interacts with the seller through one of the intermediaries. Each intermediary represents one group of consumers and provides the seller with aggregate data about the group he represents. This data corresponds to the distribution of values of the consumers in the group. For each group of consumers whose values are aggregated, the seller sets the corresponding profit-maximizing price. Thus, the payoff of a consumer in a group depends on the values of the other consumers in the group.³ This generates an incentive for high-value consumers to group with low-value consumers, which leads to a strategic tension between consumers if there are “insufficiently many” low-value consumers.

To see this tension, consider a market in which a measure 0.45 of consumers have value 1 for the product, a measure 0.35 have value 2, and the remaining measure 0.2 have value

¹“Consumer Privacy in an Era of Rapid Change, Recommendations for Businesses and Policymakers,” FTC report, March 2012.

²MiData and Salus Coop enable their members to pool and share health data, and Swash pools web surfing data from its members.

³This dependency can be thought of as a novel “network effect,” which may be positive or negative depending on the composition of the group. Network effects are widely discussed in the literature on platforms (see Rochet and Tirole, 2003 and Caillaud and Jullien, 2003).

3. Suppose first that the value 2 consumers group with the value 1 consumers. The seller optimally sets a price of 1 for this group of consumers and a price of 3 for the group of the remaining consumers, whose value is 3. If instead the value 3 consumers group with the value 1 consumers, the seller optimally sets a price of 1 for this group of consumers and a price of 2 for the group of remaining consumers, whose value is 2. Consumers with value 2 prefer the first scenario to the second one, whereas consumers with value 3 have the opposite preference.

To study how the market outcome resolves this strategic tension, we model the possibly complicated interaction between consumers and intermediaries in reduced form as a process that leads to a segmentation of consumers into groups, with the seller setting a profit-maximizing price for each group. Our main focus is on segmentations that are “stable” in an appropriate sense that we describe below. A natural benchmark based on the core is that no deviating group of consumers should be able to form that would make them all better off when facing the price the seller would set for this group (if its consumer values were aggregated). This benchmark is of limited use in our setting, however, because of its treatment of consumers who are equally happy with the original segmentation and with the proposed deviation. If such consumers can be part of a deviating group (this corresponds to the “strong core”), then no segmentation is immune to deviations if the seller finds it optimal to set a price that exceeds the lowest possible consumer value when all the consumers in the market are aggregated to a single group. These are precisely markets in which the strategic tension exists, because a small number of the consumers with the lowest value among those who purchase the product can instead profitably group with the excluded consumers (who are indifferent).⁴ If, on the other hand, the members of a deviating group must all strictly benefit from the deviation (this corresponds to the “weak core”), then every segmentation is immune to deviations, because the consumers with the lowest value in a deviating group do not benefit from the deviation.

We take the view that whether a proposed deviation is reasonable should be determined by the consumers who strictly benefit and the consumers who are strictly harmed, without assuming a-priori that indifferent consumers join one side or the other uniformly across all segmentations and deviations. We formulate three notions to capture this view. Our first notion, the “intermediate core,” stipulates that voluntarily leaving an intermediary to “break apart” an existing coalition, and then joining a new intermediary to form the deviating coalition, both involve a small disutility arising from an administrative friction.

⁴In contrast, if it is optimal for the seller to set a price equal to the lowest possible consumer value when all the consumers in the market are aggregated to a single group, then all the consumers agree that this single group is optimal, so the strategic tension does not arise and there is nothing to resolve.

The former disutility must ultimately be incurred by at least *some* of the deviating consumers from every segment that is broken apart in order to form the deviation; the latter disutility must ultimately be incurred by at least *some* of the deviating consumers who form the new coalition. Consumers who strictly benefit are willing to incur the small disutility, while indifferent consumers are not. This rules out certain deviations relative to the “strong core.”⁵

We then formulate two notions of stability based on the idea that to effectuate a move to an alternative segmentation, the consumers who initiate the move must strictly benefit from it and also be able to overcome opposition from consumers who are strictly harmed by the move.⁶ These notions also capture the idea that each consumer has control over his data and each intermediary maintains the interests of the group of consumers he represents. Our first notion of stability corresponds to a “right to leave,” (RTL) which states that if a consumer would strictly benefit from the alternative segmentation, then he cannot be prevented from leaving his current group.⁷ This captures the idea that a consumer’s consent is required to aggregate his data with those of the other members of the group. Thus, an alternative segmentation is a feasible deviation from the prevailing segmentation if any group in the prevailing segmentation that contains a consumer who is strictly harmed by the deviation also contains a consumer who strictly benefits from the deviation. A segmentation for which no such deviation exists is “RTL stable.” A segmentation that is RTL stable is immune to deviations to alternative segmentations because consumers cannot be forced to be part of a consumer group against their will.

Our second notion of stability corresponds to a “right to stay,” (RTS) which states that if no consumer in a group would strictly benefit from the move to the alternative segmentation, and some consumers in the group would be strictly harmed, then the move cannot be effectuated. This captures the idea that if a group of consumers wants the intermediary, who is acting on behalf of these consumers, to continue representing their group, then the intermediary will do so. Thus, an alternative segmentation is *not* a feasible deviation from a prevailing segmentation if one of the consumer groups in the prevailing segmentation includes consumers who are strictly harmed but no consumers who strictly benefit from the move to the alternative segmentation. A segmentation for which no such deviations exist

⁵Notice that a deviation that precludes a segmentation from being in the strong core satisfies the second requirement, because some of the deviating members strictly benefit from the deviation and are therefore willing to incur the small disutility associated with forming the deviating coalition. Our intermediate core symmetrizes this by assuming that breaking apart a segment also involves a small disutility that must be incurred by some of the deviating members in the segment.

⁶To determine which consumers are harmed, the deviation must specify the entire alternative segmentation and not only the deviating group, unlike the notions based on the core.

⁷As we discuss in Section 1.1, RTL stability is related to Balbuzanov and Kotowski (2019)’s exclusion core in exchange economies.

is “RTS stable.” A segmentation that is RTS stable is immune to deviations to alternative segmentations because consumers cannot be forced to dissolve their group due to objections from consumers in other groups (represented by other intermediaries). It is simple to show that both notions of stability are equivalent, and we refer to segmentations that satisfy them as “stable segmentations.”

Our main result characterizes stable segmentations and also shows that the set of stable segmentations coincides with the intermediate core. Our characterization also proves that stable segmentations always exist. The characterization shows that a segmentation is stable if and only if it is *efficient* and *saturated*. Efficiency means that every consumer buys the product, and saturation means that consumers in each group are not willing to accept additional consumers from groups that face higher prices because doing so increases the price their own group will face. We also show that stable segmentations are Pareto undominated, that is, there is no other segmentation that makes all consumers better off. These results suggest that data intermediaries can play a productive role in facilitating consumers’ interactions with sellers and lead to favorable market outcomes.

As an application of our characterization we study the following question. Suppose that the composition of each market segment (in terms of values) is initially determined by intermediaries before consumers learn their individual value for the product, but consumers can attempt to form new groups after they learn their value. To make this concrete, we provide a stylized example in which individuals take a test that determines their risk of contracting a certain disease, and group membership is based on the test results. The intermediaries can form “risk pools” before the consumers take the test, but the consumers may try to form new groups (with possibly different intermediaries) after taking the test. Which segmentation would arise? We formalize this as asking which stable segmentations maximize expected consumer surplus among all stable segmentations. We answer this question by constructively showing that a stable segmentation exists that maximizes average consumer surplus among all segmentations (stable or not). This stable segmentation is the *maximal equal-revenue segmentation*, identified by Bergemann, Brooks, and Morris (2015).⁸ Our result that the maximal equal-revenue segmentation is stable shows that stability does not reduce the surplus consumers can achieve even in the absence of commitment or a central planner that enforces the segmentation. This further increases the potential value of data intermediaries that represent consumers in their interaction with a seller. We also show that multiple stable segmentations may exist, and that maximizing average consumer surplus neither implies nor

⁸Bergemann, Brooks, and Morris (2015) characterized the set of average consumer-producer surplus pairs achievable across all segmentations, and did not consider stability or the question of which segmentations will arise.

is implied by stability.

Our analysis of stable segmentations may be relevant to policy discussions regarding monopolies, price discrimination, data sharing, and data intermediaries. Monopolies lead to inefficiencies, and these inefficiencies may be reduced with regulation or increased competition. Market segmentation arising from a monopolist's access to consumer data can also reduce inefficiency, but may harm consumers, as first-degree price discrimination demonstrates. Our results show that data intermediaries that facilitate consumers' interactions with sellers by aggregating the data of groups of consumers can lead to efficient and Pareto undominated outcomes for consumers. If group criteria are chosen before the consumers know their value, then the consumers can achieve the highest possible expected surplus without the help of a social planner even if the consumers and intermediaries cannot commit to maintaining the segmentation after the consumers learn their value. This indicates that policies or information intermediaries, such as data cooperatives, that facilitate consumer group formation while allowing the seller to price discriminate across groups may offer a promising alternative or complimentary tool to addressing the inefficiencies associated with monopolistic markets. Finally, while our primary motivation is markets with data intermediaries, we note that consumer groups that interact with sellers also arise in other contexts, such as employee or student unions and home-owner associations. Our analysis may also be relevant to such settings, albeit to a more limited extent.⁹

The rest of the paper is organized as follows. Section 1.1 describes the relationship of our work to the existing literature. Section 2 describes the model. Section 3 describes the core and why it is unsatisfactory in our setting. Section 4 introduces the intermediate core. Section 5 introduces our two stability notions and shows that they are equivalent. Section 6 Characterizes stable segmentations shows their equivalence to the intermediate core. Section 7 shows that stable segmentations exist and relates them to consumer-optimal segmentations. Section 9 concludes and discusses the relationship of our stability notion to existing solution concepts for non-transferable utility games. The appendix contains additional examples, proofs omitted from the text, the relationship to existing cooperative solution concepts, and an extension to a model with a continuum of values. The online appendix contains additional details and proofs omitted from the appendix.

⁹In practice, such groups benefit consumers in other ways, like increasing members' bargaining power. Our analysis shows that group formation can benefit consumers even when abstracting from these positive effects.

1.1 Related Literature

Our right-to-leave (RTL) stability is conceptually closely related to the exclusion core introduced by Balbuzanov and Kotowski (2019). They consider an exchange economy in which endowments are interpreted as a distribution of exclusion rights over assets. A coalition can “directly exclusion block” a candidate allocation of the assets with another allocation if all its members prefer the other allocation and the candidate allocation assigns to any agent who prefers the candidate allocation an asset over which member(s) of the coalition have exclusion rights. In our setting, there are no assets, and our RTL states that any member of a group of consumers can leave it (by not allowing his data to be aggregated with the rest of the group’s). But if we interpret groups as assets, then we can think of the RTL as every member of a group having the right to exclude any other member, so that leaving (or not joining) a group is equivalent to preventing the other members from “owning” the group. That is, by leaving the group, which every member of the group has the right to do, the group is dissolved (or not formed in the first place).

Peivandi and Vohra (2021) consider stability of centralized markets against deviations by coalitions of agents. They show that fragmentation of such markets is unavoidable, despite its efficiency costs, except in special circumstances. They study a bilateral trade setting, whereas we study a setting with a population of consumers and a seller. When centralized markets are fragmented, Peivandi and Vohra (2021) do not predict what the resulting segmentation looks like, whereas characterizing stable segmentations is a main focus of our paper. Another important difference is that in their setting a coalition chooses the trading mechanism, whereas in our setting each coalition of consumers faces a profit-maximizing price set by the seller.

Our work is also related to the literature that studies the role of intermediaries on market outcomes. Acemoglu et al. (2022) and Bergemann, Bonatti, and Gan (2022) study data intermediaries who collect and reveal information about consumers. Externalities arise because a consumer’s data reveals information about others. These externalities are different than the pooling tension studied in this paper where values are uncorrelated. Galperti, Levkun, and Perego (2024) and Galperti, Liu, and Perego (2024) study an intermediary that buys data from consumers and then decides how to pool them to sell to a firm. As a result, a pooling externality arises in their setting that is different than the one in our setting, where an intermediary simply aggregates the data from consumers it represents without further segmentation, and the information available to the seller is only specified by the groups that consumers form.

A recent literature on third-degree price discrimination studies consumer and producer surplus across all possible segmentations of a given market. Bergemann, Brooks, and Morris

(2015) identify the set of average producer and consumer surplus pairs that result from all segmentations of a given market. Their results also identify segmentations that maximize average consumer surplus. Haghpanah and Siegel (2023) study when a market served by a multi-product seller can be segmented in a way that is Pareto improving. Yang (2022) studies how a profit-maximizing data broker sells market segmentations to a monopolist. Ichihashi (2020), Hidir and Vellodi (2021), Braghieri (2017), and Haghpanah and Siegel (2022) consider maximum average consumer surplus when a multi-product seller offers different products in each market segment. These papers can be seen as identifying segmentations that are chosen *ex ante* by a consumer who does not know her type, because such a consumer chooses the segmentation that maximizes her expected payoff. In contrast, the main focus of our paper is the formation of market segmentations when consumers know their type. Our application to the setting in which the segmentation is chosen before consumers know their type but can try to deviate to another segmentation shows that intermediaries can help consumers achieve the highest possible consumer surplus across all segmentations even in the absence of commitment or a central planner.

More closely related to our paper is the literature that studies segmentation as the outcome of disclosure decisions by consumers who know their types. The main difference with our work is that these papers model these interactions as non-cooperative games. Ali, Lewis, and Vasserman (2023) consider voluntary disclosure of data by a single consumer, and analyze the welfare implications of various disclosure policies in both a monopolistic and a competitive environment. Sher and Vohra (2015) study a disclosure setting in which the seller can commit to the mechanism that he will use after receiving information. Kuang et al. (2022) study the formation of market segmentations as the equilibrium of a non-cooperative game in which each consumer chooses which segment to join. In their setting consumers can unilaterally move between segments, whereas in our setting, once a segmentation arises consumers cannot join an existing segment if all consumers in that segment oppose this change. These differences lead to different results: in their setting the seller's surplus in any equilibrium segmentation is equal to the seller's surplus in the unsegmented market and consumers are better off, but the outcome need not be efficient. In contrast, our stable segmentations are Pareto undominated, efficient, and increase the seller's surplus.

2 Model

A monopolistic seller faces a unit mass of consumers uniformly distributed on the unit interval $[0, 1]$. Consumers have unit demand for the monopolist's product. The value of the product for consumer $c \in [0, 1]$ is $v(c) \in V = \{v_1, \dots, v_n\} \subseteq R_{>0}$, where v is a measurable function

and v_i increases in i .¹⁰ Let μ be the Borel measure on the unit interval. The measure of consumers with value v_i is $f(v_i) = \mu(\{c : c \in [0, 1], v(c) = v_i\})$. We assume without loss of generality that $f(v_i) > 0$ for every $v_i \in V$, and normalize the seller's production cost to zero.

A *coalition* is a measurable subset $C \subseteq [0, 1]$ of consumers. Let $f^C(v_i) = \mu(\{c : c \in C, v(c) = v_i\})$ denote the measure of consumers with value v_i in coalition C . We say that *consumers with value v_i are in C* (or that C contains consumers with value v_i) if $f^C(v_i) > 0$. A price $p \in V$ is *optimal for coalition C* if it maximizes the revenue from selling the product to consumers in C , that is, for any other price $p' \in V$,

$$p \sum_{i: v_i \geq p} f^C(v_i) \geq p' \sum_{i: v_i \geq p'} f^C(v_i).$$

We restrict attention to prices in V because for any other price there exists a price in V with a higher revenue.

Consumers interact with the seller through intermediaries. We do not model the precise interaction between the consumers and the intermediaries, and assume only that it leads to a partition of the consumers into coalitions, each represented by an intermediary. Each intermediary provides the seller with the distribution of consumer values in the coalition the intermediary represents, and the seller sets an optimal price for each such coalition. Formally, a *segment* is a pair (C, p) , where C is a coalition and p is an optimal price for that coalition.¹¹ A *segmentation* S is a finite set of segments $\{(C_j, p_j)\}_{j=1, \dots, k}$ such that $C_1, \dots, C_k$ partitions the set of consumers $[0, 1]$. That is, a segmentation partitions the set of consumers into coalitions and assigns an optimal price for each coalition.

Denote by $CS(c, p) = \max\{v(c) - p, 0\}$ the surplus of consumer c who is offered the product at price p . If consumer c belongs to segment (C, p) , then her surplus is $CS(c, p)$. Given a segmentation S , denote by $p_S(c)$ the price in the unique segment that includes consumer c . Let $CS(c, S) = CS(c, p_S(c))$ denote the surplus of consumer c in segmentation S .

Consumers' preferences over segmentations may differ because the prices different consumers face vary within and across segmentations. This gives rise to strategic "pooling tension" discussed in the introduction. We investigate how this tension is resolved and which segmentations arise by abstracting from the details of the interactions between the agents and instead focusing on properties of the resulting segmentations. These properties are motivated by consumers' ownership of their data and intermediaries acting on behalf of the

¹⁰Appendix F discusses variants of our model with a continuum of possible values.

¹¹That is, the seller cannot price discriminate between the different consumers in C , and sets a price for these consumers that maximizes his revenue.

consumers they represent.

3 The Core

We first consider two natural notions motivated by the core and show that they are unsatisfactory in our setting. The idea is that a segmentation should be immune to any group of consumers attempting to increase their surplus by leaving their current intermediaries and being represented as a coalition by another intermediary. A key issue is whether consumers who are equally happy with the prevailing segmentation can be part of a deviating group.

As we now show, both notions, which take opposing and extreme stances on this issue, fail to resolve the fundamental strategic tension between consumers discussed earlier. Indeed, if all (or almost all) the consumers in the deviating group are required to strictly benefit from the deviation, which corresponds to the “weak core,” then clearly every segmentation is immune to all possible deviations, because the optimality of the price the monopolist sets for the new coalition implies that consumers with the lowest value in the new coalition have zero surplus from the deviation. Thus, this notion is too permissive to resolve the strategic tension between consumers. If, on the other hand, we require that only some of the consumers in the deviating group strictly benefit from the deviation, we obtain the “strong core,” which we will refer to as “the core.” To formalize this notion and see why it is also unsatisfactory, we first define objections.

Definition 1 (Objection) *A segment (C, p) objects to a segmentation S if $CS(c, p) \geq CS(c, S)$ for all consumers c in C , with a strict inequality for a positive measure of consumers c in C .*

Notice that an objection (C, p) to S is *not* a segment in S (otherwise all the consumers in C would be indifferent between the segment and the segmentation).

We define the core to be the set of segmentations to which there is no objection.

Definition 2 (Core) *The core is the set of segmentations to which no segment objects.*

The proposition below shows that the core is too restrictive to resolve the strategic tension between consumers. For the proposition, we say that a market is *efficient* if the seller finds it optimal to sell to all consumers when the market is unsegmented.

Definition 3 (Efficiency) *A market is efficient if price v_1 is optimal for coalition $[0, 1]$.*

If the market is efficient, then the segmentations in which every consumer faces price v_1 (including the “trivial” segmentation $\{([0, 1], v_1)\}$) are most preferred by every consumer, because v_1 is the lowest price the seller would ever offer. Because all the consumers agree on the best segmentations, there is no strategic tension. And, consequently, the core then consists of the segmentations in which every consumer faces price v_1 . On the other hand, if price v_1 is *not* optimal for the set of all consumers, then there is no segmentation in which every consumer faces price v_1 . Any consumer who faces a price higher than v_1 prefers segmentations in which she faces price v_1 . And a small measure of these consumers can join the consumers with value v_1 to face price v_1 and form an objection. Thus, the core is empty precisely when consumers disagree on their top-ranked segmentations so there is some strategic tension between them. The following proposition formalizes these observations. Its proof (and other proofs omitted from the text) are in the appendix.

Proposition 1 *If the market is inefficient, then the core is empty. Otherwise, the core consists of the segmentation $\{([0, 1], v_1)\}$ and all the other segmentations in which every consumer faces price v_1 .*

Our goal is to understand whether and how intermediaries can help resolve the strategic tension between consumers that arises when the consumers do not all agree on a top-ranked segmentation, that is, when the market is inefficient. Thus, the above core notions are not useful in our setting.

To proceed, observe that in the above notions, whether consumers who are equally happy with the prevailing segmentation can be part of a deviating group is specified in a uniform way (yes or no) across all possible segmentations and potential deviations. If equally happy consumers can always be part of a deviating group, then these consumers are effectively siding with the consumers who initiate the deviation, whereas if they cannot, then these consumers are effectively siding with the consumers who oppose the deviation. But whether a segmentation is a reasonable prediction should plausibly be determined by (possibly a subset of) the consumers who have a *strict* interest given a proposed deviation: those who would strictly benefit from the deviation and those who would strictly be harmed by it.

We introduce three notions based on this view, each focusing on a different aspect of the market interaction. Our first notion, the intermediate core, is obtained by strengthening the requirement from an objection and is an “intermediate” core notion, between “strong” and “weak.” Since it is based on objections, the intermediate core revolves around the consumers who initiate a deviation because they strictly benefit from it. Our other two notions are equivalent stability notions that reflect the balance of power between consumers who strictly support a deviation and those who strictly oppose it. A deviation in these

notions specifies the entire alternative segmentation, and not only the deviating coalition, in order to determine which consumers would strictly benefit and which consumers would strictly be harmed by the deviation.

4 The Intermediate Core

To distinguish between consumers who initiate an objection because they strictly benefit from it and consumers who are indifferent (but may be a required part of the objection), we stipulate that breaking apart an existing coalition and forming a new coalition in order to generate an objection both entail an administrative friction in the form of an arbitrarily small disutility. This (non-transferable) disutility must be incurred by at least *some* consumers in the relevant coalition. The two core notions discussed in Section 3 accommodate such disutilities for forming the new coalition but not necessarily for breaking apart existing coalitions. Indeed, because some consumers in an objecting coalition strictly benefit from the deviation, they would be willing to incur the small disutility associated with forming the new coalition.¹² But if some existing coalition that includes indifferent consumers required for the new coalition does not also include consumers from the new coalition who strictly benefit, then, because those indifferent consumers would not be willing to incur the disutility associated with breaking apart the existing coalition (in order to form the new one), the objection will fail.

This leads to our definition of a strict objection, which “symmetrizes” the small disutilities associated with breaking apart an existing coalition and with forming a new one. Formally, we say that C' *breaks apart* C if coalition C' contains some but not all of the consumers in coalition C (so $C \cap C'$ and $C \setminus C'$ both have positive measures).

Definition 4 *A segment (C', p') strictly objects to a segmentation S if*

1. (C', p') objects to S , and
2. for any segment (C, p) in S , if C' breaks apart C , then $CS(c, p') > CS(c, S)$ for a positive measure of consumers c in $C \cap C'$.

Definition 5 *The intermediate core is the set of segmentations S to which no segment strictly objects.*

¹²The weak core, which requires all the deviating coalition members to strictly benefit from the deviation, would even accommodate requiring *all* consumers in the new coalition to incur the disutility associated with forming the new coalition.

Because strict objections are objections, the intermediate core is a superset of the core.¹³ If the market is efficient, so the core consists of the segmentations in which every customer faces price v_1 (Proposition 1), then the core and the intermediate core coincide. This is because in this case the segment $([0, 1], v_1)$, which does not break apart any coalition, objects (and therefore strictly objects) to any non-core segmentation.

The following example illustrates objections, strict objections, and the intermediate core.

Example 1 (Objections, strict objections, and the intermediate core) *There are two values, 1, 2. Consumers from 0 to 0.4 have value 1, and consumers from 0.4 to 1 have value 2.*

Consider segmentation $S = \{(C_1, 1), (C_2, 2)\}$, where $C_1 = [0, 0.6)$ and $C_2 = [0.6, 1]$. One segment that objects to S is $(C'_1, 1)$, where $C'_1 = [0, 0.8)$. This objection is obtained by adding the set of consumers $[0.6, 0.8)$, with measure 0.2, from C_2 to C_1 , which leaves the low price of 1 optimal for the seller. The surplus of consumers in C_1 does not change, and the surplus of the added consumers from C_2 strictly increases. Moreover, the objection is strict because C'_1 only breaks apart C_2 , so S is not in the intermediate core.

Now consider segmentation $S' = \{(C'_1, 1), (C'_2, 2)\}$, where $C'_1 = [0, 0.8)$ and $C'_2 = [0.8, 1]$. Adding any positive measure of consumers from C'_2 to C'_1 necessarily increases the optimal price in the first segment, so such a coalition would not be an objection because the value 2 consumers in C'_1 would be strictly harmed. However, segmentation S' is not in the core because segment $(C''_1, 1)$, where $C''_1 = [0, 0.4] \cup [0.8, 1]$, objects to S' : value 1 consumers in C''_1 are indifferent, and value 2 consumers in C''_1 strictly prefer $(C''_1, 1)$ to S' , where they face price 2. However, $(C''_1, 1)$ is not a strict objection: C''_1 breaks apart C'_1 , but none of the consumers in $C'_1 \cap C''_1$ strictly prefer $(C''_1, 1)$ to S' . In fact, S' is in the intermediate core. This is because for any objection (C'''_1, p) to S' , C'''_1 breaks apart C'_1 (otherwise $C'_1 \subset C'''_1$ or $C'''_1 \subset C'_2$, so $p = 2$ and no consumer in C'''_1 strictly prefers (C'''_1, p) to S'); and consumers in C'_1 face in S' the lowest possible price of 1.

5 Stable Segmentations

We now introduce our two notions of stability, which are concerned with the balance of power between consumers who would strictly benefit and consumers who would strictly be harmed by a deviation. To identify these consumers, a deviation in these notions specifies the entire alternative segmentation. Our first notion of stability, right-to-leave stability, is motivated by the idea that a consumer's consent is required to aggregate his data with those of the

¹³However, as we show in the example below, not every objection is a strict objection.

other members of his coalition, so he cannot be forced to be in a coalition against his will. Thus, consumers who would strictly benefit from moving to an alternative segmentation can overcome the opposition (by consumers who would be strictly harmed by the move) and dissolve the entire prevailing segmentation if any coalition in the prevailing segmentation that contains consumers who are strictly harmed by the move also contains consumers who strictly benefit from the move.¹⁴ To formalize this, we first define right-to-leave blocking. For the definition, it is helpful to think of segmentation S as the prevailing segmentation and segmentation S' as an alternative segmentation.

Definition 6 (Right-To-Leave (RTL) Blocking) *A segmentation S' right-to-leave (RTL) blocks a segmentation S if a positive measure of consumers c strictly prefer S' to S and every segment in S that contains a positive measure of consumers who strictly prefer S to S' also contains a positive measure of consumers who strictly prefer S' to S .¹⁵*

A segmentation is RTL stable if no segmentation RTL blocks it.

Definition 7 (Right-To-Leave (RTL) Stability) *A segmentation is right-to-leave (RTL) stable if it is not RTL blocked by any other segmentation.*

RTL stability requires that for any alternative segmentation that some consumers strictly prefer, there is a segment in the prevailing segmentation that contains some consumers who strictly prefer the prevailing segmentation (and therefore oppose the move) but no consumers who strictly prefer the alternative segmentation (and therefore support the move). If that is the case, then not every segment in the prevailing segmentation can be dissolved, so the balance of power favors the opposing consumers. The move then fails and the segmentation is stable. This interpretation highlights the second key idea that underlies our notions of stability, namely that intermediaries represent the interests of their customers. So, if some customers in a coalition strictly oppose a move and no customer in the coalition strictly supports it, then the coalition remains intact and the move fails.

¹⁴This is similar to the idea that underlies Balbuzanov and Kotowski (2019)’s exclusion core in the context of an exchange economy. In their setting, given a candidate allocation of the assets, a coalition can “directly exclusion block” this allocation with another allocation if all its members prefer the other allocation and the candidate allocation assigns to any agent who prefers the candidate allocation an asset over which member(s) of the coalition have exclusion rights. Because in our setting every member of a coalition can leave it, thereby dissolving the coalition, if we think of a coalition as an asset, then every member of a coalition can “exclude” its other members from using this “asset.”

¹⁵The definition may seem to imply that the consumers who strictly prefer S' to S are “dictating” how the other consumers will be rearranged in S' . But the proof of our main result will show that if S' RTL blocks S , then there is another segmentation that also RTL blocks S and which does not “rearrange” the other consumers (those who do not strictly prefer the other segmentation to S). Footnote 19 elaborates on this point.

Our second notion of stability, right-to-stay stability, formalizes this idea. To define this notion, we first introduce right-to-stay blocking. Recall that a segment objects to a segmentation if all the consumers in the segment prefer the price in the segment to the price they face in the segmentation, and for some of these consumers the preference is strict. For the definition of right-to-stay blocking, it is again helpful to think of segmentation S as the prevailing segmentation and segmentation S' as an alternative segmentation.

Definition 8 (Right-To-Stay (RTS) Blocking) *A segmentation S right-to-stay (RTS) blocks a segmentation S' if there exists a segment (C, p) in S that objects to S' .*

For the definition of right-to-stay stability, we say that segmentations S and S' are *equivalent* if almost all consumers face the same price in the two segmentations, that is, for almost all consumers c in $[0, 1]$, c is in a segment with price p in segmentation S if and only if c is in a segment with price p in segmentation S' .¹⁶

Definition 9 (Right-To-Stay (RTS) Stability) *A segmentation is right-to-stay (RTS) stable if it RTS blocks any non-equivalent segmentation.*

RTS stability focuses on a coalition's "right to stay," which allows the prevailing segmentation to block deviations to other segmentations, whereas RTL stability focuses on consumers' "right to leave," which allows other segmentations to block the prevailing segmentation. Despite these differences in their motivation, the two notions are equivalent, as the following result states.

Lemma 1 *A segmentation is RTS stable if and only if it is RTL stable.*

Because RTS and RTL stability are equivalent, we refer to segmentations that satisfy these notions as "stable segmentations." The following example illustrates RTL blocking, RTS blocking, and stability.

Example 2 (RTL blocking, RTS blocking, and stability) *Consider Example 1 and the segmentation $S' = \{(C'_1, 1), (C'_2, 2)\}$, where $C'_1 = [0, 0.8)$ and $C'_2 = [0.8, 1]$. As discussed earlier, segment $(C''_1, 1)$, where $C''_1 = [0, 0.4] \cup [0.8, 1]$, objects to S' . But notice that segment $(C'_1, 1)$ in S' objects to any segmentation that contains $(C''_1, 1)$. This is because in any such segmentation, any consumer not in C''_1 has value 2 and faces price 2. Thus, S' RTS blocks*

¹⁶Notice that two equivalent segmentations need not be identical when viewed as two sets of segments. For example, given a segmentation, if we replace a segment (C, p) with two segments $(C \setminus C', p)$ and (C', p) for some subset C' of C (so that price p is optimal for both coalitions C' and $C \setminus C'$), then we obtain a new segmentation that is not identical to the original segmentation but is equivalent to it.

any segmentation that contains $(C_1'', 1)$. In addition, any segmentation S'' such that some consumers strictly prefer S'' to S' does not RTL block S' . This is because some consumers in C_1' strictly prefer S' to S'' (otherwise all the consumers with value 2 in C_1' face in S'' price 1, so all the consumers in C_2' must face in S'' price 2 and no consumers strictly prefer S'' to S') but no consumers in C_1' strictly prefer S'' to S' (since consumers in C_1' face in S' the lowest possible price of 1).

In fact, S' is stable. This is because $(C_1'', 1)$ objects to any non-equivalent segmentation, so S' RTS blocks any non-equivalent segmentation. Indeed, because consumers in C_1' face in S' the lowest possible price of 1, all of them weakly prefer S' to any other segmentation. And if a segmentation S'' is not equivalent to S' , then some of the value 2 consumers in C_1' must face in S'' price 2 and therefore strictly prefer S' to S'' .

Notice that segmentation S' from Example 1 and Example 2 is both in the intermediate core and is stable. Our main result, to which we now turn, shows that this is not a coincidence.

6 Main Result: Characterization and Equivalence

Our main result characterizes the intermediate core and the set of stable segmentations and shows that they coincide. For the characterization, we start by introducing the notion of a *canonical* segmentation. A segmentation is canonical if no two segments in it have the same price.¹⁷ Each segmentation S is equivalent to a unique canonical segmentation, which we call the *induced canonical segmentation* of S . The induced canonical segmentation is obtained by merging segments that have the same price into a single segment with that price.¹⁸

Our characterization says that a segmentation is stable if and only if its induced canonical segmentation satisfies two properties: efficiency and saturation. A segmentation is *efficient* if all consumers buy the product, that is, for any segment (C, p) in the segmentation, the price p is equal to the lowest value $\underline{v}(C) := \min\{v : f^C(v) > 0\}$ of consumers in C . A segmentation is *saturated* if for any segment (C, p) in the segmentation, whenever we add consumers to coalition C from a segment with a price strictly higher than p , price p is suboptimally low for this larger coalition. That is, for any two segments (C, p) and (C', p') in the segmentation with $p < p'$, and any positive-measure set $C'' \subseteq C'$ of consumers, any optimal price for coalition $C \cup C''$ is strictly higher than p .

¹⁷Bergemann, Brooks, and Morris (2015) refer to these as *direct* segmentations.

¹⁸Suppose that price p is optimal for disjoint coalitions $C^1, \dots, C^k$, that is, $p \sum_{i:v_i \geq p} f^{C^j}(v_i) \geq p' \sum_{i:v_i \geq p'} f^{C^j}(v_i)$ for all C^j and p' . Summing over all j , and letting $C = \bigcup_j C^j$, we have $p \sum_{i:v_i \geq p} f^C(v_i) \geq p' \sum_{i:v_i \geq p'} f^C(v_i)$ for all p' , so price p is optimal for coalition C .

The following lemma shows that saturation can be expressed more succinctly by looking at the set of prices that are optimal for different segments.

Lemma 2 *A segmentation is saturated if and only if for any two segments (C, p) and (C', p') in the segmentation with $p < p'$, there exists a price $\hat{p}$ that is optimal for C such that $p < \hat{p} \leq \underline{v}(C')$.*

Proof. If such a $\hat{p}$ exists, then by adding consumers from C' to C , all of whose values are at least $\underline{v}(C')$, and therefore at least $\hat{p}$, the revenue from price $\hat{p}$ increases more than the revenue from price p . And because both p and $\hat{p}$ are optimal for C , p (and any price lower than p) is not optimal when we add these consumers. Conversely, if no such $\hat{p}$ exists, then p is the highest optimal price for C that does not exceed $\underline{v}(C')$, so if we add a small measure of consumers with value $\underline{v}(C')$ from C' to C , price p remains optimal for C . ■

We now state and prove our main result.

Theorem 1 *For any segmentation S , the following statements are equivalent*

1. *S is in the intermediate core.*
2. *S is stable.*
3. *The canonical segmentation of S is efficient and saturated.*

Proof. We first show that if S is stable, then it is in the intermediate core by proving the contrapositive. To see this, suppose that S is not in the intermediate core, so there is a strict objection (C', p') to S . We can then construct a segmentation S' that RTL blocks S . Start with S' being the empty set. Add (C', p') to S' . For any segment (C, p) in S such that $C \cap C'$ is empty (has zero measure), add (C, p) to S' . For any segment (C, p) such that C is broken apart by C' , add segment $(C \setminus C', p'')$ to S' , where p'' is any optimal price for $C \setminus C'$. To see that S' RTL blocks S , consider three cases. If $C \subseteq C'$, then because (C', p') objects to S , the consumers in C weakly prefer S' to S . If $C \cap C'$ is empty, then (C, p) is a segment in S' , so the consumers in C are indifferent between S and S' . If C is broken apart by C' , then because (C', p') is a strict objection to S , some consumers in $C \cap C'$ strictly prefer S' to S . Finally, note that some consumers strictly prefer S' to S , so S' RTL blocks S .¹⁹

¹⁹Notice that S' RTL blocks S without the need for consumers in C' to arbitrarily “rearrange” remaining consumers. In particular, for any segment (C, p) in S that has some consumers not in C' , we have that $(C \setminus C', p'')$ is a segment in S' . This formalizes the claim made in Footnote 15 that RTL blocking does not require the consumers who strictly benefit from the deviation to be able to arbitrarily arrange the other consumers. Instead, they simply leave their current coalitions.

We have shown that statement (2) in the theorem implies statement (1). We will show that statement (1) implies statement (3), and statement (3) implies statement (2). This last implication is the least straightforward part of the proof.

We first show that statement (1) implies statement (3). To see that any segmentation in the intermediate core has an efficient canonical segmentation, consider a segmentation S' with an induced canonical segmentation S . Suppose that S is inefficient. We will construct a strict objection to S' . Because S is inefficient, so is S' , so there is a segment (C', p') in S' with $p' > \underline{v}(C')$. Consider a coalition $\bar{C} \subseteq C'$ that consists of all the consumers in C' with values strictly lower than p' , in addition to a positive measure of the highest value consumers in C' that is small enough that any optimal price for $\bar{C}$ is strictly lower than p' . Denote by $p < p'$ an optimal price for $\bar{C}$, so $(\bar{C}, p)$ is a segment. Observe that p' remains optimal for $C' \setminus \bar{C}$. Indeed, removing from C' consumers with values strictly lower than p' , who do not purchase the product, does not change the revenue from p' ; and removing from C' some consumers with the highest value in C' can only lower the optimal price, but p' is already the lowest value of consumers in C' after removing the consumers with values lower than p' , so p' remains optimal. We argue that $(\bar{C}, p)$ is a strict objection to S' . To see this, first notice that $(\bar{C}, p)$ is an objection to S' : because $p < p'$, the consumers in $\bar{C}$ weakly prefer segment $(\bar{C}, p)$ to S' , and the preference is strict for the consumers with the highest value in C' that are in $\bar{C}$. Further, notice that C' is the only coalition in S' that is broken apart by $\bar{C}$, and the consumers in $\bar{C} \cap C'$ who have the highest value in C' strictly prefer $(\bar{C}, p)$ to S' , so $(\bar{C}, p)$ is a strict objection to S' .

To see that any segmentation in the intermediate core has a saturated canonical segmentation, consider a segmentation S' with an induced canonical segmentation S . Suppose that S is not saturated. If S is inefficient, then the argument above implies that S' is not in the intermediate core. Suppose that S is efficient, which together with non-saturation implies (by Lemma 2) that there are two segments $(C, \underline{v}(C))$ and $(C', \underline{v}(C'))$ in S with $\underline{v}(C) < \underline{v}(C')$ such that no $\hat{p}$ with $\underline{v}(C) < \hat{p} \leq \underline{v}(C')$ is optimal for C . In particular, if we add a positive-measure set $C'' \subset C'$ of consumers with value $\underline{v}(C')$ to C , price $\underline{v}(C)$ remains optimal provided that the measure of C'' is sufficiently small. Let C'' be such a set that contains a positive measure of consumers with value $\underline{v}(C')$ from *every* segment in S' in which the price is $\underline{v}(C')$ (recall that S' need not be canonical). Let $\bar{C} = C \cup C''$ and consider the segment $(\bar{C}, \underline{v}(C))$.

We argue that $(\bar{C}, \underline{v}(C))$ is a strict objection to S' . Segment $(\bar{C}, \underline{v}(C))$ objects to S' because the consumers in C'' strictly prefer it to the segmentation, and the consumers in $C = \bar{C} \setminus C''$ are indifferent between $(\bar{C}, \underline{v}(C))$ and S' . Now consider which coalitions in S' are broken apart by $\bar{C}$. Coalitions in segments with a price different from $\underline{v}(C)$ or $\underline{v}(C')$ do not intersect with $\bar{C}$. The coalition of a segment with price $\underline{v}(C)$ in S' is a subset of $\bar{C}$, so these

coalitions are not broken apart by $\bar{C}$. The only coalitions that *are* broken apart by $\bar{C}$ are those in which the price is $\underline{v}(C')$. By construction, from any such coalition there are some consumers in $C'' \subseteq \bar{C}$, who strictly prefer $(\bar{C}, \underline{v}(C))$ to S' . So $(\bar{C}, \underline{v}(C))$ is a strict objection to S' .

We now show that statement (3) implies statement (2). Consider a segmentation S' with an induced canonical segmentation S that is efficient and saturated. Let $\bar{S}$ be a segmentation that is not RTS blocked by S' . We will show that $\bar{S}$ is equivalent to S' . Since the canonical representation of $\bar{S}$ is also not RTS blocked by S' and is equivalent to S' if and only if $\bar{S}$ is equivalent to S' , we suppose without loss of generality that $\bar{S}$ is canonical. Write the two canonical segmentations as $S = \{(C_1, v_1), \dots, (C_n, v_n)\}$ and $\bar{S} = \{(\bar{C}_1, v_1), \dots, (\bar{C}_n, v_n)\}$, where for each i either C_i is empty or $\underline{v}(C_i) = v_i$ (because S is efficient), and each $\bar{C}_i$ may be empty. We will show by induction that $C_i = \bar{C}_i$ for all i , which will prove that $\bar{S}$ is equivalent to S' . (For the rest of this proof, $C_i = \bar{C}_i$ is in the “almost all” sense, that is, the measure of consumers in C_i but not in $\bar{C}_i$ is zero, and the measure of consumers in $\bar{C}_i$ but not in C_i is zero).

Suppose that $C_j = \bar{C}_j$ for all $j < i$ (the basis of the induction is $i = 1$). We show that $C_i = \bar{C}_i$. If $i = n$, then we are done because S and $\bar{S}$ partition the same set $[0, 1]$ of consumers. Suppose that $i < n$. Since $C_j = \bar{C}_j$ for all $j < i$, a consumer faces a price $p \geq v_i$ in S if and only if she faces a price $p' \geq v_i$ in $\bar{S}$ (up to a measure zero of consumers). In particular, consumers in C_i face a price of at least v_i in $\bar{S}$. Consumers in C_i with values higher than v_i must be in $\bar{C}_i$, otherwise these consumers face a price strictly higher than v_i in $\bar{S}$, so any segment in S' that contains some of these consumers objects to $\bar{S}$ (and thus S' RTS blocks $\bar{S}$): the consumers in this segment are in C_i so face price v_i in S' (because S is the canonical representation of S'), and in $\bar{S}$ the consumers in this segment face prices no lower than v_i (by the claim earlier in the paragraph). So C_i and $\bar{C}_i$ are identical (up to a measure zero of consumers), except that $\bar{C}_i$ possibly does not contain some consumers of value v_i from C_i and may contain some consumers from coalitions $C_{i+1}, \dots, C_n$, all of whom have values strictly higher than v_i (because S is efficient). But, as we now argue, if C_i and $\bar{C}_i$ are not identical, then the fact that S is saturated contradicts the fact that v_i is optimal for $\bar{C}_i$ (so $(\bar{C}_i, v_i)$ is not a segment). To see this, suppose first that $\bar{C}_i$ does not contain some consumers of value v_i from C_i . Since S is saturated and $i < n$, by Lemma 2 some price $p > v_i$ is optimal for C_i and p is lower than the value of all consumers in coalitions $C_{i+1}, \dots, C_n$. Removing from C_i some consumers with value v_i reduces the revenue from price v_i but not the revenue from price p , which makes price v_i sub-optimal. Then, if needed, adding to C_i consumers from coalitions $C_{i+1}, \dots, C_n$, all of whose values are at least p , to obtain $\bar{C}_i$ makes price v_i even worse (weakly) relative to price p , so v_i is not optimal for $\bar{C}_i$. Now suppose that

$\bar{C}_i$ differs from C_i only because $\bar{C}_i$ contains some consumers from coalitions $C_{i+1}, \dots, C_n$, all of whom have value strictly higher than v_i . Adding these consumers to C_i makes price v_i sub-optimal because S is saturated, so v_i is not optimal for $\bar{C}_i$. ■

The proof of Theorem 1 in fact shows that if a segmentation is stable, then it is efficient and saturated, regardless of whether it is canonical. But the other direction of the proof, that efficiency and saturation imply stability, relies on the segmentation being canonical. Example 6 in Online Appendix G describes a non-canonical segmentation that is efficient and saturated but not stable (and therefore, by Theorem 1, is also not in the intermediate core and does not satisfy the intermediate notion of stability).

6.1 Discussion of the Main Result

The equivalence between stability and the intermediate core implies that if a segmentation is not stable, then it has a strict objection. Moreover, in Appendix C we show that if a segmentation is not stable, there is a finite sequence of strict objections that takes us from the original segmentation to one that is stable.

We can also use Theorem 1 to formulate a different approach to the question of which segmentations will arise. Instead of starting from the intermediate core or stability, we can specify desirable properties of a segmentation and ask which solution concepts they correspond to. Efficiency and saturation are natural properties for a segmentation that arises when intermediaries represent groups of consumers in their interaction with the seller. As the simpler parts of the proof of Theorem 1 show, if a canonical segmentation is not efficient or not saturated, then simple deviations exist. Theorem 1 shows that if such deviations are ruled out, that is, if efficiency and saturation hold, then in fact all strong objections are ruled out and the segmentation RTS blocks any non-equivalent segmentation, that is, we obtain the intermediate core and stability.

7 Existence, Consumer-optimal Stable Segmentations, and Welfare Implications

We show that stable segmentations always exist by using the characterization from Theorem 1 to construct a stable segmentation. We start with an example that demonstrates the construction.

Example 3 (Maximal equal-revenue segmentation) *There are three values, 1, 2, 3. Consumers from 0 to $\frac{1}{3}$ have value 1, those from $\frac{1}{3}$ to $\frac{1}{2}$ have value 2, and those from $\frac{1}{2}$ to 1 have value 3, with measures $\frac{1}{3}, \frac{1}{6}, \frac{1}{2}$, respectively, as shown in Figure 1.*

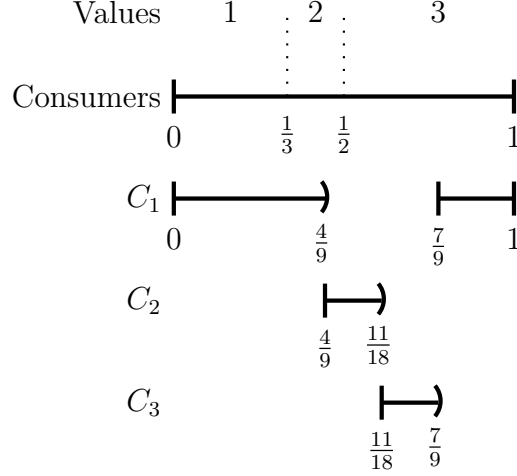


Figure 1: Example 3.

Consider the segmentation $S = \{(C_1, 1), (C_2, 2), (C_3, 3)\}$, where $C_1 = [0, \frac{4}{9}) \cup [\frac{7}{9}, 1]$, $C_2 = [\frac{4}{9}, \frac{11}{18})$, and $C_3 = [\frac{11}{18}, \frac{7}{9})$, shown in Figure 1. Coalition C_1 is the largest “equal-revenue” coalition that includes all values. That is, the measures $\frac{1}{3}, \frac{1}{9}, \frac{2}{9}$ of the three values in coalition C_1 are such that prices 1, 2, and 3 are all optimal, and C_1 is the largest such coalition because it contains all the consumers with value 1.²⁰ Segment $(C_1, 1)$ is efficient. Consumers not in C_1 have value either 2 or 3, so adding them to C_1 makes price 1 no longer optimal.

Having put all the consumers with value 1 in segment C_1 , we define the rest of the segmentation recursively to guarantee efficiency and saturation. The values of the remaining consumers are 2 and 3, and the measures of these consumers are $\frac{1}{18}$ and $\frac{5}{18}$, respectively. Coalition C_2 , in which values 2 and 3 have measures $\frac{1}{18}$ and $\frac{2}{18}$, is the largest coalition for which prices 2 and 3 are both optimal. The segment $(C_2, 2)$ is efficient, and adding any of the remaining consumers, all of whom have value 3, increases the optimal price. The last segment is $(C_3, 3)$, which is efficient. Thus, segmentation S is efficient and saturated. Because it is also canonical, it is stable by Theorem 1.

We now formally define the maximal equal-revenue segmentation (MERS). Let $\bar{F}^C(v_i)$ be the cumulative measure of consumers with values v_i or higher in coalition C . If $v_i \bar{F}^C(v_i) = v_j \bar{F}^C(v_j)$, then prices v_i and v_j generate the same revenue for coalition C . Coalition C is an equal-revenue coalition if all consumer values in the coalition generate the same revenue, that is, $v_i \bar{F}^C(v_i)$ is the same for all v_i with $f^C(v_i) > 0$. The MERS is defined recursively. The first coalition, C_1 , is the largest equal-revenue coalition that includes all the values. To construct C_1 , let λ_1 be the eventual revenue in coalition C_1 from each of the values, that is,

²⁰Any two coalitions C and C' for which all three prices are optimal are proportional, that is, $f^C(v) = \alpha f^{C'}(v)$ for some $\alpha > 0$ and every value v , so the largest such coalition is well-defined.

$\lambda_1 = v_i \bar{F}^{C_1}(v_i)$ for all v_i in V . Recalling that $f^C(v_i)$ is the measure of consumers with value v_i in coalition C , and $f(v_i)$ is the overall measure of consumers with value v_i , we have that $f(v_i) \geq f^{C_1}(v_i) = \bar{F}^{C_1}(v_i) - \bar{F}^{C_1}(v_{i+1}) = \lambda_1(\frac{1}{v_i} - \frac{1}{v_{i+1}})$ for all v_i , where $\frac{1}{v_{n+1}} \equiv 0$. Therefore, the highest value that λ_1 can take is such that $f^{C_1}(v_i) = f(v_i)$ for some i . That is, λ_1 is the smallest value such that $\lambda_1(\frac{1}{v_i} - \frac{1}{v_{i+1}}) = f(v_i)$ for some i . Denote the index of this value by i_1 , so $\lambda_1(\frac{1}{v_{i_1}} - \frac{1}{v_{i_1+1}}) = f(v_{i_1})$. Then, more succinctly, we define C_1 by letting

$$\lambda_1 = \min_{v_i \in V} \frac{f(v_i)}{\frac{1}{v_i} - \frac{1}{v_{i+1}}} = \frac{f(v_{i_1})}{\frac{1}{v_{i_1}} - \frac{1}{v_{i_1+1}}}, \quad (1)$$

and letting $\bar{F}^{C_1}(v_i) = \lambda_1/v_i$ for all v_i .

Coalition C_1 contains all the consumers with value v_{i_1} , and adding a positive measure of consumers with other values to C_1 makes price v_{i_1} sub-optimal. Therefore coalition C_1 cannot be any larger and still be an equal-revenue coalition. The first segment in the MERS is (C_1, v_1) , and the rest of the segmentation is defined recursively, where coalition C_j in the j 'th segment is the largest equal-revenue coalition that includes all the values that remain after removing the consumers in $C_1, \dots, C_{j-1}$, that is, $\{v_i : f^{C_j}(v_i) > 0\} = \{v_i : f^{[0,1] \setminus \cup_{j' < j} C_{j'}}(v_i) > 0\}$, and the price in the j 'th segment is $\min\{v_i : f^{C_j}(v_i) > 0\}$. This process ends because in each step the number of remaining values decreases by at least 1.

The MERS is not necessarily canonical. For example, if the first equal-revenue coalition C_1 exhausts some value other than v_1 , then the second coalition, C_2 , will also include consumers with value v_1 , so the MERS will contain two segments, (C_1, v_1) and (C_2, v_1) , that have the same price. By Theorem 1, to establish that the MERS is stable, we need show that its induced canonical segmentation is efficient and saturated.

Proposition 2 *The maximal equal-revenue segmentation (MERS) is stable.*

Bergemann, Brooks, and Morris (2015), who first introduced the MERS, showed that the MERS maximizes average consumer surplus across all segmentations (stable or not). They also showed that segmentations other than the MERS maximize average consumer surplus. This raises the question of the relationship between stability and maximizing average consumer surplus and whether the MERS is the only stable segmentation. Example 4 and Example 5 in Appendix E show that stability is neither necessary nor sufficient for maximizing average consumer surplus.

Even though not all stable segmentations necessarily maximize average consumer surplus, they satisfy another desirable welfare property. We say that segmentation S' *Pareto dominates* segmentation S if $CS(c, S') \geq CS(c, S)$ for all consumers c in $[0, 1]$, with a strict

inequality for a positive measure of consumers. A segmentation S is *Pareto undominated* if no segmentation Pareto dominates S .

Lemma 3 *Stable segmentations are Pareto undominated. Pareto undominated segmentations are efficient.*

Proof. If S' Pareto dominates S , then no segment in S objects to S' , and S' is not equivalent to S . Therefore, S is not stable.

For the second statement, suppose that S is inefficient, so there is a segment (C, p) in S with $p > \underline{v}(C)$. Consider a coalition $\bar{C} \subseteq C$ that consists of the consumers in C with values strictly lower than p and a positive measure of the highest value consumers in C that is small enough that any optimal price for $\bar{C}$ is strictly lower than p . Denote by $p' < p$ an optimal price for $\bar{C}$, so $(\bar{C}, p')$ is a segment. Observe that p remains optimal for $C \setminus \bar{C}$. Indeed, removing from C consumers with values strictly lower than p , who do not purchase the product, does not change the revenue from p ; and removing from C some consumers with the highest value in C can only lower the optimal price, but p is already the lowest value of consumers in C after removing the consumers with values lower than p , so p remains optimal. Now consider segmentation $\bar{S}$ obtained from segmentation S by replacing segment (C, p) with the two segments $(C \setminus \bar{C}, p)$ and $(\bar{C}, p')$. The consumers in $\bar{C}$ with the highest value in C have a strictly higher surplus in $\bar{S}$ than in S , and all the other consumers in C have a weakly higher surplus in $\bar{S}$ than in S . Thus, $\bar{S}$ Pareto dominates S . ■

Any segmentation that maximizes average consumer surplus is Pareto undominated. And because some segmentations that maximize average consumer surplus are not stable (see Example 4 in Appendix E), there are Pareto undominated segmentations that are not stable.

8 Application: Implementing a Consumer-Optimal Segmentation

Proposition 2 helps address the following question. Suppose that the intermediaries, who endeavor to represent consumers' interests, initially determine the composition of each market segment (in terms of values) before consumers learn their individual value for the product. After consumers learn their values they can attempt to deviate from the segmentation by forming new coalitions if the segmentation is not stable. Which segmentation would arise and what surplus will it achieve?

Before they learn their value, all consumers rank segmentations by the average consumer surplus they generate. Thus, a segmentation that maximizes average consumer surplus across

all segmentations (stable or not) is preferred by all consumers. And if such a segmentation is stable, then consumers will not deviate from it after they learn their value. Therefore, because the MERS maximizes average consumer surplus across all segmentations and is stable, intermediaries that have consumers' best interest in mind can help consumers achieve the maximal average surplus by specifying the MERS (or another equivalent stable segmentation) before consumers learn their value, even in the absence of commitment or a central planner that enforces the segmentation after consumers learn their value.

As an illustrative example, consider a genetically determined disease whose probability of impacting a given individual varies across individuals. Suppose that this probability can be verifiably determined by genetic testing, and suppose for simplicity that such testing is costless. Also for simplicity, suppose that the disutility in monetary terms of having the disease is the same across individuals. A monopolistic drug maker produces a drug at zero marginal cost that has no side effects and is guaranteed to prevent the disease if taken. Before learning their individual probability of having the disease, all individuals agree that grouping individuals by their probability in a way that implements the MERS achieves the highest ex-ante surplus for them. Thus, intermediaries that represent individuals' best interests will agree to form "risk pools" that will be populated by the individuals based on their test results.²¹ After the individuals learn their test result, a group of individuals may try to deviate and form its own risk pool (with a new or existing intermediary). But since the MERS is stable, no such deviations will succeed, so the individuals can obtain the maximal expected surplus across all segmentations.

Applied to this setting, Example 3 corresponds to a population with three risk levels, where the expected disutility of half of the population is 1, the expected disutility of a sixth of the population is 2, and the expected disutility of the remaining third of the population is 3. Before learning their idiosyncratic disutility (derived from their probability of being impacted by the disease), the consumers would agree that the best option is to form three risk pools, which correspond to the three segments $(C_1, 1)$, $(C_2, 2)$, and $(C_3, 3)$ in Example 3. This maximizes their expected consumer surplus across all segmentations. And once they learn their risk, no group of consumers would be able to form a strong objection, and every alternative, non-equivalent segmentation would be blocked. Thus, intermediaries would be able to help the consumers achieve their optimal ex-ante outcome.

²¹A lottery, perhaps in the form of a priority list, can be used if several pools include individuals with the same test results.

9 Conclusions

We study the role that data-collecting intermediaries may play in shaping market outcomes when consumers interact with a monopolistic seller. Without intermediaries, a seller with access to detailed consumer data would implement first-degree price discrimination, which achieves efficiency but eliminates all consumer surplus. If a social planner who wishes to maximize average consumer surplus controlled the seller’s access to consumer data, he could help achieve the highest possible average consumer surplus across all segmentations, identified by Bergemann, Brooks, and Morris (2015). Our analysis shows that intermediaries that represent groups of consumers and act on their behalf can achieve efficient and (for consumers) Pareto undominated outcomes, and may increase average consumer surplus up to that highest possible value. If the intermediaries can decide on group-membership criteria before consumers know their value, then, similarly to the case with a social planner, the highest possible average consumer surplus across all segmentations can be achieved without commitment or a social planner. This suggests that data-collecting intermediaries that represent consumers may serve as a possible alternative or addition to standard anti-trust regulation. Our analysis is a preliminary step in an intriguing field of possibilities brought about by data-collecting intermediaries, demonstrates the beneficial role that such intermediaries can play, and enlarges the scope of the discussion about consumer privacy and price discrimination that regulators are having.

Part of our methodological contribution is to introduce three notions that capture “reasonable” market outcomes in our setting from three different perspectives. Our main result shows that the three notions are equivalent and characterizes them, and we refer to segmentations that satisfy them as stable. Figure 2 illustrates the relationship between the various notions we discuss in the paper. Our stability notion refines several solution concepts for non-transferable utility (NTU) games applied to our setting. In Online Appendix H, we show that a stable segmentation together with its equivalent segmentations form a Morgenstern and Von Neumann (1953) stable set, a Harsanyi (1974) farsighted stable set, a Ray and Vohra (2015) farsighted stable set, and a Ray and Vohra (2019) maximal farsighted stable set. We point out that stable sets and farsighted stable sets do not always exist in NTU games. And when they do exist they may necessarily include multiple, non-equivalent surplus vectors or coalitions of players. Our results show that in our market game, “singleton” (up to equivalent segmentations) stable and farsighted stable sets always exist. In particular, for any deviation from a stable segmentation S to a non-equivalent segmentation S' , there exists a path of “credible” and “maximal” (in the sense of Ray and Vohra, 2019) segmentations that leads back from S' to S . This provides another justification for our notion of stability.

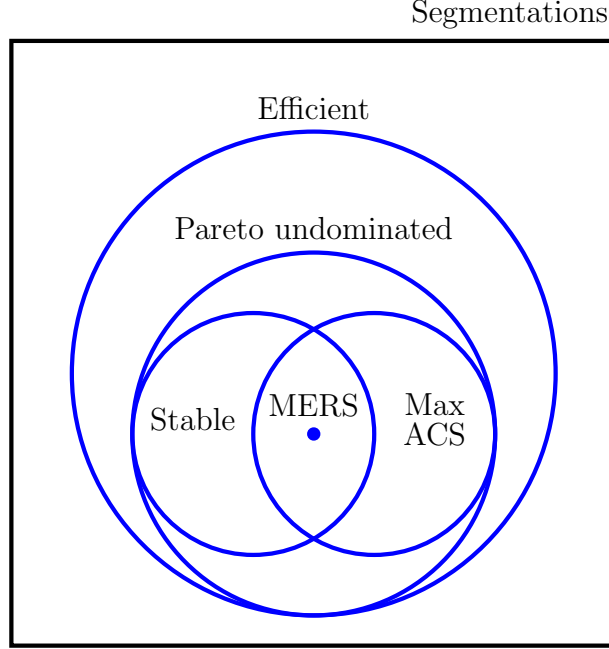


Figure 2: Summary of the relation between different notions, where “Max ACS” refers to segmentations that maximize average consumer surplus.

Appendix F discusses a variant of our model with a continuum of values, and shows that our results extend to this setting. We also discuss the effect of requiring strict improvements for all members of an objecting coalition, and show that it does not make the core a more interesting solution concept because the core becomes too large and includes all efficient segmentations; strict improvements also make stability too demanding and less interesting by not allowing objections that include indifferent consumers who “go along” with members of their existing segment who strictly prefer the existing segment.

Appendix

A Proof of Proposition 1

Proof. For the first claim, suppose that the market is inefficient, so v_1 is not optimal for the set $[0, 1]$ of all consumers. Then any segmentation includes a segment with a price strictly higher than v_1 . Otherwise, the price in every segment is equal to v_1 (because it cannot be less than v_1), so price v_1 remains optimal for the single coalition $[0, 1]$, which contradicts the assumption that v_1 is not optimal for the set of all consumers.

Now, take a segmentation S and a segment (C, p) in it with $p > v_1$. Because p is optimal

for C , C contains a positive measure of consumers with value p . Consider a coalition C' that consists of a positive measure of consumers from C with value p and a positive measure of consumers with value v_1 (from any segment). If $f^{C'}(p)$ is small enough relative to $f^{C'}(v_1)$, then price v_1 is optimal for C' , so (C', v_1) is a segment. The surplus of value p consumers in C' is $p - v_1 > 0$, whereas their surplus in S is zero. The surplus of consumers with value v_1 in any segment is zero. Therefore, segment (C', v_1) objects to S , so S is not in the core. Since S was an arbitrary segmentation, the core is empty.

For the second claim, suppose that the market is efficient. Then $([0, 1], v_1)$ is a segment and there is no objection to the segmentation $\{([0, 1], v_1)\}$ because in any segment the price is at least v_1 . For the same reason, any segmentation in which every consumer faces price v_1 is also in the core. Now consider a segmentation S in which the price p in some segment exceeds v_1 . Because p is optimal for the coalition in the segment, the coalition contains a positive measure of consumers with value p , who strictly prefer price v_1 . And because the price in any segment is at least v_1 , the segment $([0, 1], v_1)$ objects to S , so S is not in the core. ■

B Proof of Lemma 1

Proof. Suppose a segmentation S is not RTL stable, so there exists a segmentation S' that RTL blocks S . RTL blocking requires that any segment in S that has some consumers who strictly prefer S to S' also has some consumers who strictly prefer S' to S , so no segment in S objects to S' and therefore S does not RTS block S' and S is not RTS stable. Conversely, suppose S is not RTS stable so there is some non-equivalent segmentation S' that is not RTS blocked by S . Consider two cases, first where there exists some segment (C, p) in S that contains some consumers who strictly prefer S to S' , and second where no such segment exists. In the first case, because S does not RTS block S' , some consumers in any such segment must prefer S' to S , so S' RTL blocks S . In the second case, because the two segmentations are non-equivalent, some consumers prefer S' to S and no consumer prefers S to S' , so S' RTL blocks S . ■

C Supporting arguments for Section 6.1

In this section we give supporting arguments for the claims made in Section 6.1 that if a segmentation is not stable, then there is a finite sequence of strict objections that takes us from the original segmentation to one that is stable. This process follows the construction in the proof of Theorem 1 and consists of two phases. In the first phase, we convert the

segmentation to an efficient one, and in the second phase, to one whose canonical representation is also saturated. In the first phase, for each segment (C, p) such that $p > \underline{v}(C)$, we proceed exactly as in the proof of Theorem 1 and construct a strict objection that contains consumers in C with value lower than p and a small measure of consumers with the highest value in C . Because we do this once for each inefficient segment, the first phase ends after a finite number of steps. In the second phase, as long as there is a segment in the canonical representation of the segmentation that can accommodate some consumers form a segment with a higher price, take such a segment (C, p) with the lowest price, and let (C', p') be the segment with the next lowest price. Merge all the segments with price p in the segmentation (recall that the segmentation might not be canonical). Starting with $\epsilon = 0$, add a subset C'' of C' with proportional measures, $f^{C''}(v) = \epsilon f^{C'}(v)$ for all v , to C and increase ϵ continuously until either increasing ϵ increases the optimal price for $C \cup C''$ or C' becomes empty (in which case we keep adding consumers from the segment with the next lowest price). Because we are adding consumers proportionally, the optimal price for $C' \setminus C''$ does not change and efficiency is preserved. The second phase also ends in a finite number of steps because there are at most as many steps as there are values.

D Proof of Proposition 2

Proof. The price in each segment of the MERS is equal to the lowest consumer value in the segment, so the segmentation is efficient and the same is true for its induced canonical segmentation. It remains to show that the induced canonical segmentation is saturated.

By construction of the MERS, for any two segments (C_i, v_i) and (C_j, v_j) with $i < j$, the set of consumer values in C_j is a subset of that in C_i . Since coalition C_j contains consumers with value v_j ($f^{C_j}(v_j) > 0$), so does coalition C_i . Because C_i is an equal-revenue segmentation, price v_j is optimal for coalition C_i . Consider the segment with price v_i in the induced canonical segmentation. By definition of the induced canonical segmentation, the coalition in this segment is the union of all the coalitions with price v_i in the MERS. Price v_j is optimal for each of these coalitions, as argued above, and is therefore optimal for the union of these coalitions. Thus, the induced canonical segmentation is saturated by Lemma 2. ■

E Stability versus Maximizing Average Consumer Surplus

The following two examples show that stability is neither necessary nor sufficient for maximizing average consumer surplus.

Example 4 (A segmentation that maximizes average consumer surplus and is not stable) Consider again the setting from Example 3, with three values, 1, 2, 3, where consumers from 0 to $\frac{1}{3}$ have value 1, those from $\frac{1}{3}$ to $\frac{1}{2}$ have value 2, and those from $\frac{1}{2}$ to 1 have value 3, with measures $\frac{1}{3}, \frac{1}{6}, \frac{1}{2}$, respectively. Consider segmentation $S = \{(C_1, 1), (C_2, 2)\}$ with coalitions $C_1 = [0, \frac{1}{3}] \cup [\frac{5}{6}, 1]$ and $C_2 = (\frac{1}{3}, \frac{5}{6})$.

Coalition C_1 contains all the value 1 consumers and some value 3 consumers in a proportion that makes prices 1 and 3 optimal. Coalition C_2 contains the remaining consumers, whose proportions are such that prices 2 and 3 are optimal. Segmentation S maximizes average consumer surplus across all segmentations.²²

But the segmentation is not stable. Since it is canonical and efficient, to show that it is not stable, we show that it is not saturated. Indeed, adding a small measure $\epsilon > 0$ of value 2 consumers to C_1 does not make price 1 sub-optimal: price 2 is not optimal for C_1 , so if ϵ is small enough, price 2 remains sub-optimal, and the addition increases the revenue from price 1 but does not change the revenue from price 3.

Example 5 (A stable segmentation that does not maximize average consumer surplus) There are three values, 1, 2, 3. Consumers from 0 to $\frac{1}{3}$ have value 1, those from $\frac{1}{3}$ to $\frac{2}{3}$ have value 2, and those from $\frac{2}{3}$ to 1 have value 3, with measures $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$, respectively, as shown in Figure 3.

Consider the segmentation $S = \{(C_1, 1), (C_2, 3)\}$ where $C_1 = [0, \frac{2}{3})$ and $C_2 = [\frac{2}{3}, 1]$ shown in Figure 3. Coalition C_1 consists of the consumers with values 1 and 2. Coalition C_2 consists of the consumers with value 3. This segmentation is canonical, efficient, and saturated. It is therefore stable.

The average consumer surplus of this segmentation is $\frac{1}{3}$, whereas the maximum average consumer surplus is $\frac{2}{3}$.²³ For some intuition, it is illuminating to study the marginal improvement in the average consumer surplus of S obtained by swapping the same measure of

²²The segmentation is efficient so it maximizes total surplus. It also minimizes the seller's revenue across all segmentations. This is because price 3, which is optimal for the set of all consumers, is also optimal for each coalition.

²³To find maximum average consumer surplus, we can consider the MERS. The MERS is $\{(C_1'', 1), (C_2'', 2)\}$, where $(f^{C_1''}(1), f^{C_1''}(2), f^{C_1''}(3)) = (\frac{1}{3}, \frac{1}{9}, \frac{2}{9})$ and $(f^{C_2''}(1), f^{C_2''}(2), f^{C_2''}(3)) = (0, \frac{2}{9}, \frac{1}{9})$. The surplus of value 2 and 3 consumers is 1 and 2 in the first segment, and the surplus of value 3 consumers is 1 in the second segment. The average consumer surplus is therefore $\frac{1}{9} \cdot 1 + \frac{2}{9} \cdot 2 + \frac{1}{9} \cdot 1 = \frac{2}{3}$.

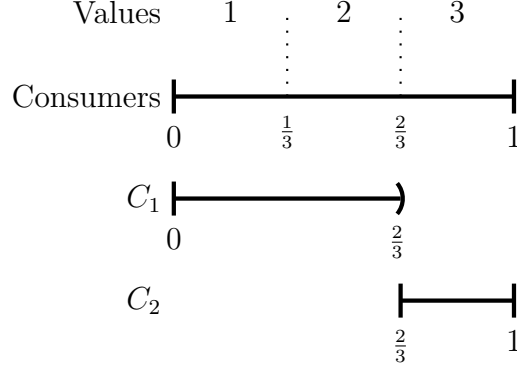


Figure 3: Example 5.

value 2 and 3 consumers. For this, consider a coalition C'_1 obtained from C_1 by removing measure ϵ of value 2 consumers and adding a measure ϵ of value 3 consumers, and a coalition C'_2 that contains the remaining consumers. If $\epsilon > 0$ is small enough, price 1 is optimal for C'_1 and price 3 is optimal for C'_2 , so $S' = \{(C'_1, 1), (C'_2, 3)\}$ is a segmentation. To compare the average consumer surplus of S and S' , it suffices to consider the swapped consumers. Each value 2 consumer loses 1 unit of surplus: their surplus is 1 in S and 0 in S' . Each value 3 consumer gains 2 units of surplus: their surplus is 0 in S and 2 in S' .²⁴

Example 5 can be generalized to give a construction of a stable segmentation that is different from the MERS. Put all consumers with value v_1 in the first coalition, and continually add consumers with the lowest remaining value to the first coalition until some price v_i other than v_1 also becomes optimal. This forms the first coalition, C_1 . The first segment is (C, v_1) . Repeat this process with the remaining consumers (the last segment may have only one optimal price). The resulting segmentation is canonical, efficient, and saturated. Saturation follows because given a segment (C, v_j) so constructed, a value $v_k > v_j$ becomes optimal for coalition C only when we have already added all the available consumers with values lower than v_k to C , so the value of any consumer in a segment with a higher price is at least v_k , and adding such consumers to C makes price v_j sub-optimal. This segmentation is also typically different from the MERS because the first segment does not generally include all values.²⁵

²⁴Notice that the change in the offered price is 2 for the value 2 consumers (from 1 to 3) and -2 for the value 3 consumers (from 3 to 1). But even though this change has the same absolute value, the surplus change for the value 3 consumers is higher than for the value 2 consumers because value 2 consumers do not buy the product at a price higher than 2 (so increasing the price they face from 2 to 3 does not change their surplus.)

²⁵This alternative stable segmentation is related to the greedy algorithm of Ali, Lewis, and Vasserman (2023), adapted to the case of a finite number of values. However, whereas our construction starts from the lowest type, theirs starts from the highest type. As a result, their segmentation may not be stable. For example, suppose that there are three possible values, 1, 2, 4, each with measure $\frac{1}{3}$. The greedy algorithm

F Continuum of Values

One modeling assumption we make is that the number of possible consumer values is finite. We consider a variant of our setting with a continuum of values and show that our main results extend: the core is empty unless the unsegmented market is efficient, and stability is characterized by efficiency and saturation.

In this setting with a continuum of values, we also study what happens if we strengthen the definition of an objection to require that almost all consumers in a segment strictly benefit from deviating to the segment. This reduces the number of deviations relative to the setting that allows for objections with indifferences, which makes the core a less demanding solution concept. With this stronger form of objections, the core becomes too permissive in the sense that it contains all finite efficient segmentations. We also study what stability looks like with this stronger form of objections. Because we require a given segmentation to veto any deviation in a stronger sense, we should expect stability to become more demanding. In fact, stability becomes too demanding in the sense that no stable segmentations exist. Our view is that indifferences are consistent with an intuitive notion of coalitional IR and stability: a consumer who is indifferent could reasonably agree to passively go along with the objecting votes of her current coalition members who are strictly harmed by a proposed deviation. In contrast, when defining the core, it may be reasonable to require strict improvements: to be willing to actively deviate from a segmentation, consumers in the coalition should strictly benefit.

In the variant of our setting we consider there is a unit mass of consumers whose values are distributed on an interval $[\underline{v}, \bar{v}]$, $0 < \underline{v} < \bar{v} < \infty$ according to a distribution F with a derivative that is bounded from above and away from 0. Each consumer $c \in [\underline{v}, \bar{v}]$ is identified by her unique value for the product. A coalition C consists of a finite union of sub-intervals of $[\underline{v}, \bar{v}]$. A segment (C, p) consists of a coalition C and a price p that is optimal (revenue-maximizing) when the values are drawn according to F conditional on being in C . The assumption that the derivative of F is bounded from above and away from 0 means that there exists some $\delta > 0$ such that for any consumer c , price $p = c$ is uniquely optimal for the set of consumers $[c, c + \epsilon]$ for any $\epsilon \leq \delta$.

We focus on finite segmentations. A (finite) segmentation is a finite set of segments $\{(C_j, p_j)\}_{j=1, \dots, k}$ such that $C_1, \dots, C_k$ partition the set of all consumers $[\underline{v}, \bar{v}]$.²⁶ Let $CS(c, p)$

from Ali, Lewis, and Vasserman (2023) puts consumers with values 2 and 4 in one segment with price 2, and those with value 1 in another segment with price 1. This segmentation is not saturated because we can add consumers from the segment with price 2 to the one with price 1 without increasing the price in the latter segment. Our construction puts consumers with values 1 and 2 in one segment with price 1, and those with value 4 in another segment with price 4. This segmentation is stable.

²⁶In this formulation each consumer has a unique value. An alternative formulation, inspired by the

denote the surplus of consumer c from being offered price p , and $CS(c, S)$ the surplus of this consumer in segmentation S .

We first study the core and stability according to the notion of objection used throughout the paper, requiring that all consumers in the segment weakly, and some of them strictly, prefer the segment to the segmentation. Recall that stability is defined using a notion of equivalence. Here we say that two segmentations are equivalent if, for any segment (C, p) in one segmentation, there exists a segment (C', p) in the other segmentations such that C and C' are almost identical, that is, both $C \setminus C'$ and $C' \setminus C$ have zero measure. Our characterization is unchanged: the core is empty unless the unsegmented market is efficient, and stability is characterized by efficiency and saturation.²⁷

Proposition 3 *If the unsegmented market is efficient, that is, price $\underline{v}$ is optimal for the set $[\underline{v}, \bar{v}]$ of all consumers, then the core consists of the segmentation $\{([\underline{v}, \bar{v}], \underline{v})\}$. Otherwise, the core is empty. A segmentation is stable if and only if it is efficient and saturated.*

Proof. If the unsegmented market is efficient, then in the segmentation $\{([\underline{v}, \bar{v}], \underline{v})\}$ all consumers are offered the lowest price $\underline{v}$ and so there is no objection to the segmentation. Further, in any segmentation in the core, all consumers must be offered price $\underline{v}$, otherwise the segment that contains all consumers together with price $\underline{v}$ is an objection, so the segmentation $\{([\underline{v}, \bar{v}], \underline{v})\}$ is the unique segmentation in the core.

Suppose the unsegmented market is inefficient. Then, in any segmentation, there must exist a segment with price p strictly higher than $\underline{v}$. Take a small measure δ of consumers with value at least p from that segment, and add them to coalition $[\underline{v}, \underline{v} + \epsilon)$ to form a new coalition C . For small enough ϵ , price $\underline{v}$ is uniquely optimal for $[\underline{v}, \underline{v} + \epsilon)$, so if δ is small enough, price $\underline{v}$ is also optimal for coalition C so $(C, \underline{v})$ is an objection.

We now turn to stability. If a segmentation S is inefficient, then it contains some segment (C, p) in which the price p is higher than the lowest value. Replace (C, p) with two segments: a new segment (C', p') with consumers whose values are lower than p , and another segment containing the remaining consumers $(C \setminus C', p)$. We must have $p' < p$ because all consumers in C' have value at most p . Also C' must contain a positive measure of consumers with

information design literature, would be as follows. An unsegmented market is a distribution F over values $[\underline{v}, \bar{v}]$. A segment is a pair (G, p) , where p is an optimal price when values are distributed according to G . A segmentation is a finite set of segments $\{(G_j, p_j)\}_{j=1, \dots, k}$ and a distribution over the segments given by probabilities $\alpha_1, \dots, \alpha_k$ satisfying Bayes-plausibility, $F = \sum_j \alpha_j G_j$. In this formulation, it is possible for multiple segments to contain consumers of some value. The issue now is that it is ambiguous to talk about a segment (G, p) objecting to a segmentation S , because there is no way to keep track of *which* segments in the original segmentation the consumers in the segment are coming from. We leave an appropriate formalization of such a model for future work.

²⁷In this setting with a continuum of values, any segmentation is canonical because any two segments in any segmentation have different prices. As a result, any two segmentations are non-equivalent.

value strictly higher than p' (otherwise the revenue is zero). Call the resulting segmentation S' . Notice that (C, p) does not object to S' because some of the consumers, those in (C', p') with values above p' , strictly prefer S' to S . So S is not stable.

Suppose a segmentation S is efficient but not saturated. Saturation means there are two segments $(C, \underline{v}(C))$ and $(C', \underline{v}(C'))$ with prices $\underline{v}(C) < \underline{v}(C')$ in S such that we can add some consumers from C' to C without increasing the price in the first segment. Call the resulting segmentation S' . The segment $(C, \underline{v}(C))$ does not object to S' because the consumers in it are indifferent. The segment $(C', \underline{v}(C'))$ does not object to S' because some of its consumers, those added to the segment with the lower price, strictly prefer S' to S . So S is not stable.

Suppose that a segmentation $S = \{(C_1, p_1), \dots, (C_k, p_k)\}$, $p_1 < \dots < p_k$ is efficient and saturated. Suppose there is some segmentation S' that is not blocked by S . Let $C'_1, \dots, C'_k$ be the coalitions in S' belonging to segments with prices $p_1, \dots, p_k$, respectively (these coalitions might be empty). Because S is efficient, p_1 is equal to the lowest possible value $\underline{v}$. Therefore, if C_1 contains a positive measure of consumers that are not in C'_1 , then (C_1, p_1) objects to S' . By saturation, C'_1 cannot be a superset of C_1 , therefore we must have $C'_1 = C_1$ (in the almost all sense). Now again because S is efficient, p_2 is equal to the lowest value among consumers that are not in C_1 . A similar argument implies that we must therefore have $C'_2 = C_2$. A recursive argument implies that S and S' must be equivalent. ■

We now consider a stronger notion of an objection in which almost all consumers are required to strictly prefer the segment.

Definition 10 (Strong objection) *A segment (C, p) strongly objects to a segmentation S if $CS(c, p) > CS(c, S)$ for almost all consumers in C .*

Definition 11 (Weak core) *The weak core is the set of segmentations S to which no segment strongly objects.*

Definition 12 (Strong stability) *A segmentation is strongly stable if it contains a strong objection to any non-equivalent segmentation.*

We first demonstrate the main ideas with an example. Suppose values are uniformly distributed on $[1, 3]$. For any $\delta \leq 1$, price v is optimal for a coalition $[v, v + \delta]$.²⁸ So $S = \{([1, 1 + \delta], 1), ([1 + \delta, 1 + 2\delta], 1 + \delta), \dots, ([1 + k\delta, 3], 1 + k\delta)\}$ is a segmentation and is efficient.

We argue that S is in the core for any $\delta \leq 1$. So suppose $\delta \leq 1$ and there exists some strong objection (C, p) to S . Because consumers in segment $([1, 1 + \delta], 1)$ are offered

²⁸The revenue from price p is proportional to $p(v + \delta - p)$, and its derivative is $v + \delta - 2p \leq \delta - v$ because $p \leq v$, and $\delta - v \leq 0$ because $\delta \leq 1 \leq v$.

the lowest possible price, they cannot be a part of any strong objection, so $[1, 1 + \delta) \cap C$ has measure zero. This in turn implies that $p \geq 1 + \delta$. But then consumers in segment $([1 + \delta, 1 + 2\delta), 1 + \delta)$ cannot be a part of a strong objection either. An inductive argument shows that a strong objection does not exist.

We show below that the weak core is the set of all efficient segmentations. Our interpretation is that the weak core is a weak solution concept because there are many efficient segmentations. Recall that because the derivative of F is bounded above and away from 0, there exists $\delta > 0$ such that for any consumer c , price c is revenue-maximizing for the set of consumers $[c, c + \epsilon]$ for any $\epsilon \leq \delta$. Now we construct a class of efficient segmentations. Starting from $c = \underline{v}$, choose an arbitrary $\epsilon \leq c + \delta$ and let $C = [c, c + \epsilon]$ and add segment (C, c) to the segmentation. Repeat until all consumers are in the segmentation.

Proposition 4 *A segmentation is in the weak core if and only if it is efficient.*

Proof. Suppose first that a segmentation S is inefficient. Therefore, there exists a segment (C, p) in S such that p is strictly higher than $\underline{v}(C) = \inf\{v(c) | c \in C\}$. Let p' be the optimal price for the set $C \cap \{v | v \leq p\}$ (which is a positive-measure set because $p > \underline{v}(C)$). Notice that p' is also optimal for the set $C' := C \cap \{v | p' \leq v \leq p\}$. The segment (C', p') is a strong objection to S because all these consumers get zero surplus in S but almost all of them get a positive surplus in (C', p') .

Now suppose that a segmentation S is not in the core, so it has an objection (C, p) . Because S is finite and the coalition in each segment in S consists of a finite union of intervals, there is an $\epsilon > 0$ such that all consumers in $(p, p + \epsilon)$ belong to a single segment in S , say (C', p') . Because price p is optimal for C , a positive measure of consumers in $(p, p + \epsilon)$ must be in C (otherwise we can increase the price without decreasing revenue). So because (C, p) is a strong objection to S , the consumers in $(p, p + \epsilon) \cap C$ must strictly prefer (C, p) to the segmentation, and it must be that $p < p'$. But this implies that p' is less than the value of a positive measure of consumers in C' , so (C', p') is inefficient and therefore S is also inefficient.²⁹ ■

With strong objections, stable segmentations do not exist.

Proposition 5 *There exists no strongly stable segmentation.*

Proof. Consider any segment (C, p) in any segmentation S . Let C' be a subset of C consisting of consumers with value at least $c' > p$ (with c' chosen so that C' has positive measure). Any optimal price p' for C' is strictly higher than p . Because we removed consumers with

²⁹Notice that this argument does not require the objecting coalition C to consist of a finite union of intervals.

the highest values from C , any optimal price p'' for $C \setminus C'$ is at most p . Now replace (C, p) with (C', p') and $(C \setminus C', p'')$, and call the resulting segmentation S' . The two segmentations are not equivalent because $p' \neq p$. But (C, p) is not a strong objection to S' because the consumers in $C \setminus C'$ weakly prefer S' to S . ■

References

- Acemoglu, Daron, Ali Makhdoumi, Azarakhsh Malekian, and Asu Ozdaglar. 2022. “Too much data: Prices and inefficiencies in data markets.” *American Economic Journal: Microeconomics* 14 (4):218–256.
- Ali, S Nageeb, Greg Lewis, and Shoshana Vasserman. 2023. “Voluntary Disclosure and Personalized Pricing.” *The Review of Economic Studies* 90 (2):538–571.
- Balbuzaanov, Ivan and Maciej H. Kotowski. 2019. “Endowments, exclusion, and exchange.” *Econometrica* 87 (5):1663–1692.
- Bergemann, Dirk, Alessandro Bonatti, and Tan Gan. 2022. “The economics of social data.” *The RAND Journal of Economics* 53 (2):263–296.
- Bergemann, Dirk, Benjamin Brooks, and Stephen Morris. 2015. “The limits of price discrimination.” *American Economic Review* 105 (3):921–57.
- Braghieri, Luca. 2017. “Targeted Advertising and Price Discrimination in Intermediated Online Markets.” *Working paper* .
- Caillaud, Bernard and Bruno Jullien. 2003. “Chicken & egg: Competition among intermediation service providers.” *RAND journal of Economics* :309–328.
- Galperti, Simone, Aleksandr Levkun, and Jacopo Perego. 2024. “The value of data records.” *Review of Economic Studies* 91 (2):1007–1038.
- Galperti, Simone, Tianhao Liu, and Jacopo Perego. 2024. “Competitive markets for personal data.” In *ACM EC*. ACM Economics and Computation.
- Haghpanah, Nima and Ron Siegel. 2022. “The Limits of Multiproduct Price Discrimination.” *American Economic Review: Insights* 4 (4):443–58.
- . 2023. “Pareto-improving segmentation of multiproduct markets.” *Journal of Political Economy* 131 (6):1546–1575.

- Harsanyi, John C. 1974. “An equilibrium-point interpretation of stable sets and a proposed alternative definition.” *Management science* 20 (11):1472–1495.
- Hidir, Sinem and Nikhil Vellodi. 2021. “Privacy, personalization, and price discrimination.” *Journal of the European Economic Association* 19 (2):1342–1363.
- Ichihashi, Shota. 2020. “Online Privacy and Information Disclosure by Consumers.” *American Economic Review* 110 (2):569–95.
- Kuang, Zhonghong, Sanxi Li, Yi Liu, and Yang Yu. 2022. “Stable Market Segmentation against Price Discrimination.” *Working Paper* .
- Lanier, Jaron and E Glen Weyl. 2018. “A blueprint for a better digital society.” *Harvard Business Review* 26:2–18.
- Morgenstern, Oskar and John Von Neumann. 1953. *Theory of games and economic behavior*. Princeton university press.
- Peivandi, Ahmad and Rakesh V Vohra. 2021. “Instability of Centralized Markets.” *Econometrica* 89 (1):163–179.
- Ray, Debraj and Rajiv Vohra. 2015. “The farsighted stable set.” *Econometrica* 83 (3):977–1011.
- . 2019. “Maximality in the farsighted stable set.” *Econometrica* 87 (5):1763–1779.
- Rochet, Jean-Charles and Jean Tirole. 2003. “Platform competition in two-sided markets.” *Journal of the european economic association* 1 (4):990–1029.
- Sher, Itai and Rakesh Vohra. 2015. “Price discrimination through communication.” *Theoretical Economics* 10 (2):597–648.
- Yang, Kai Hao. 2022. “Selling consumer data for profit: Optimal market-segmentation design and its consequences.” *American Economic Review* 112 (4):1364–1393.

ONLINE APPENDIX

G An Example from Section 6

The following example describes a non-canonical segmentation that is efficient and saturated but not stable. Thus, to verify the stability of a segmentation, efficiency and saturation must be checked for its induced canonical segmentation.

Example 6 *There are three values, 1, 2, 3, with measures 0.5, 0.25, 0.25, respectively, as shown in Figure 4.*

Consider the segmentation $S = \{(C_1, 1), (C_2, 1), (C_3, 3)\}$ with coalitions C_1, C_2, C_3 , shown in Figure 4.

For coalition C_1 , prices 1 and 2 are optimal. For coalition C_2 , prices 1 and 3 are optimal. Adding consumers from segment $(C_3, 3)$ to either segment $(C_1, 1)$ or $(C_2, 1)$ necessarily increases the optimal price in the latter segments. Thus, the segmentation is saturated. The segmentation is also clearly efficient.

For coalition $C_1 \cup C_2 = [0, \frac{7}{8})$, however, price 1 is the unique optimal price, so the segmentation $\{(C_1 \cup C_2, 1), (C_3, 3)\}$ is efficient but not saturated: we can add some consumers from C_3 , all of whom have value 3, to $C_1 \cup C_2$ without changing the optimal price. To see that S is not stable, note that segmentation $S' = \{([0, \frac{7}{8} + \epsilon), 1), ([\frac{7}{8} + \epsilon, 1], 3)\}$ for some small $\epsilon > 0$ Pareto dominates S , so S is not stable by Lemma 3.

H Relation to Existing Cooperative Concepts

We here relate our notions of stability to several existing solution concepts for NTU games, including various stable set notions. The cooperative solution concepts we discuss, which are typically defined for games with a finite number of players, require minor adjustments because we have a continuum of consumers. With these adjustments, we show that stability is a strict refinement of all these concepts. Stability therefore inherits the justifications for these concepts that often rely on farsighted behavior of agents, but has the additional (myopic) justification based on coalitional IR.

H.1 Stable Set

Our notion of stability is related to the notion of a *stable set* from Morgenstern and Von Neumann (1953). The stable set is defined for any cooperative game; we present its application

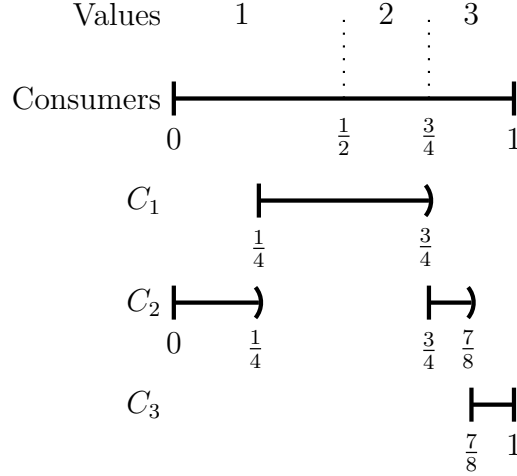


Figure 4: Example 6.

to our game. Notice that whereas the stability notion of Morgenstern and Von Neumann (1953), stated below, is a property of a *set* of segmentations, our notion of stability is a property of a single segmentation.

Definition 13 (Stable set, Morgenstern and Von Neumann, 1953) *A set of segmentations $\mathcal{S}$ is a stable set if it satisfies the following two properties:*

1. *Internal Stability: For any $S \in \mathcal{S}$, no $S' \in \mathcal{S}$ blocks S .*
2. *External Stability: For any $S \notin \mathcal{S}$, some $S' \in \mathcal{S}$ blocks S .*

If a segmentation S is stable, then the set of all segmentations that are equivalent to S is a stable set. This is easy to see: internal stability is trivially satisfied because a segmentation does not block an equivalent segmentation, and external stability is satisfied by definition of stability. Because stable segmentations always exist, stable sets exist in our setting. This is noteworthy because stable sets do not exist for some cooperative games. Moreover, even when stable sets exist, they may necessarily contain multiple elements. In contrast, our characterization of stable sets in the Online Appendix (Section H.1.1) shows that any stable set in our setting contains an essentially unique element in the sense that it consists of all the segmentations that are equivalent to some segmentation S .

Perhaps surprisingly, our characterization of stable sets shows that the set of segmentations that are equivalent to a segmentation S may be a stable set even if S is not stable. This is because stability requires that a *single* segmentation block any other non-equivalent segmentation; for the set of segmentations that are equivalent to S to be a stable set, on the other hand, requires that any segmentation that is not equivalent to S be blocked by *some*

segmentation that is equivalent to S . It may be that S does not block S' but a segmentation that is equivalent to S does. To see this, suppose that S is canonical but not stable, and consider another segmentation S' that is not blocked by S . Take a segment (C, p) in S . Since (C, p) does not object to S' , C may contain some consumers who prefer S to S' and some consumers who prefer S' to S . If coalition $C' \subseteq C$ is such that (C', p) and $(C \setminus C', p)$ are segments and we replace (C, p) with (C', p) and $(C \setminus C', p)$, it could be that (C', p) objects to S' , yielding a segmentation that is equivalent to S and blocks S' . We provide an example of this and additional details below.

H.1.1 Supporting Analysis for the Stable Set

We first provide an example where a segmentation S does not block S' but a segmentation that is equivalent to S does.

Example 7 *There are three values, 1, 2, 3, with measures $\frac{6}{21}, \frac{4}{21}, \frac{11}{21}$, respectively, as shown in Figure 5, and a segmentation $S = \{(C_1, 1), (C_2, 2)\}$ with $C_1 = [0, \frac{6}{21}) \cup [\frac{18}{21}, 1]$ and $C_2 = [\frac{6}{21}, \frac{18}{21})$.*

Segmentation S is not stable because it is not saturated. This is because we can add some consumers with value 2 from C_2 to C_1 without increasing the optimal price $p = 1$ in the first segment. It is also easy to see directly that S is not stable. For example, segmentation $S' = \{(C'_1, 1), (C'_2, 3)\}$ with coalitions $C'_1 = [0, \frac{7}{21}) \cup [\frac{18}{21}, 1]$ and $C'_2 = [\frac{7}{21}, \frac{18}{21})$ shown in Figure 5 is not blocked by S . Segment $(C_1, 1)$ in S does not object to S' because all consumers in C_1 are indifferent between the two segmentations. Segment $(C_2, 2)$ does not object to S' because the value 2 consumers who join the first segment in S' strictly prefer S' to S . However, segmentation S' is blocked by segmentation $S'' = \{(C''_1, 1), (C''_2, 2), (C'''_2, 2)\}$ with coalitions $C''_1 = [0, \frac{6}{21}) \cup [\frac{18}{21}, 1]$, $C''_2 = [\frac{6}{21}, \frac{7}{21}) \cup [\frac{16}{21}, \frac{18}{21})$, and $C'''_2 = [\frac{7}{21}, \frac{16}{21})$, which is equivalent to S . In particular, segment $(C'''_2, 2)$ objects to S' because the consumers in C'''_2 face price 2 in S'' and price 3 in S' .

The proposition below characterizes stable sets and shows that in this example the set of all segmentations that are equivalent to S is in fact a stable set. To state the proposition, we first define two weak notions of objection and blocking.

Definition 14 (Weak Objections) *A segment (C, p) weakly objects to a segmentation S if $CS(c, p) > CS(c, S)$ for a positive measure of consumers c in C , and, for every price p' that is optimal for C , $CS(c, p) \geq CS(c, S)$ for a positive measure of consumers c in C whose value is p' .*

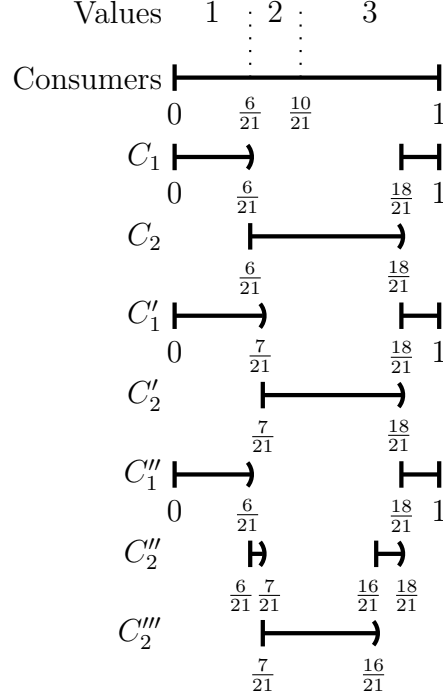


Figure 5: Example 7

Any objection is also a weak objection. To see this, observe that both objections and weak objections require that some consumers strictly prefer the segment to the segmentation. But objections also require that all consumers in the segment weakly prefer the segment. Weak objections do not require this for consumers whose value is not an optimal price for the segment. And for values that are optimal prices for the segment, only some consumers with such values are required to prefer the segment. We now define the corresponding notion of weak blocking.

Definition 15 (Weak Blocking) *A segmentation S weakly blocks a segmentation S' if there exists a segment (C, p) in S that weakly objects to S' .*

Segmentation S in Example 7 weakly blocks (but does not block) segmentation S' because segment $(C_2, 2)$ weakly objects to S' : consumers with value 3 in C_2 strictly prefer the segment to S' , consumers with value 2 in $C_2 \cap C'_2$ weakly prefer the segment to S' , and 2 is an optimal price for C_2 .³⁰

The following proposition characterizes the stable sets.

³⁰ $(C_2, 2)$ does not object to S' because consumers in $C_2 \setminus C'_2$ face a price of 2 in $(C_2, 2)$ and a price of 1 in S' .

Proposition 6 *A set of segmentations $\mathcal{S}$ is a stable set if and only if it comprises all the segmentations that are equivalent to some canonical segmentation S that weakly blocks any segmentation S' that is not equivalent to S .*

Proof. To see the necessity of these conditions, consider a stable set $\mathcal{S}$ of segmentations. We first show that any segmentation in $\mathcal{S}$ is Pareto undominated. Suppose for contradiction that a segmentation S in $\mathcal{S}$ is Pareto dominated by another segmentation S' . If S' is in $\mathcal{S}$, then internal stability is violated because S' blocks S . If S' is not in $\mathcal{S}$, then, by external stability, there is a segmentation S'' in $\mathcal{S}$ that blocks S' . But then S'' also blocks S , which violates internal stability. Pareto undominance implies that any segmentation in $\mathcal{S}$ is efficient.

We now show that any two segmentations in $\mathcal{S}$ are equivalent. Suppose for contradiction that segmentations S_1 and S_2 in $\mathcal{S}$ are not equivalent. Their induced canonical segmentations S'_1 and S'_2 are also not equivalent, so there is a price p and segments (C_1, p) in S'_1 and (C_2, p) in S'_2 with $C_1 \neq C_2$ (in the “almost all” sense), where C_1 or C_2 may be empty. Suppose without loss of generality that p is the lowest such price, so any consumer in C_1 is either in C_2 or in a segment of S'_2 with a higher price, and similarly any consumer in C_2 is either in C_1 or in a segment of S'_1 with a higher price (up to a set of consumers of measure 0). Because $C_1 \neq C_2$, either $C_1 \setminus C_2$ or $C_2 \setminus C_1$ has positive measure. Suppose without loss of generality that $C_1 \setminus C_2$ has positive measure.

First observe that $C_1 \setminus C_2$ cannot contain a positive measure of consumers with value p . Indeed, such consumers would be in segments of S'_2 with prices strictly higher than p , so S'_2 would not be efficient, contradicting the efficiency of S_2 . Therefore, $C_1 \setminus C_2$ contains a positive measure of consumers with values higher than p .

Consider any segment (C'', p) in S_1 that contains some such consumers, that is, $C'' \cap (C_1 \setminus C_2)$ has positive measure. Because $C'' \subseteq C_1$, the consumers in C'' face prices no lower than p in S_2 , and the consumers in $C'' \cap (C_1 \setminus C_2)$ face prices strictly higher than p in S_2 . So S_1 blocks S_2 , which contradicts internal stability.

We have established that $\mathcal{S}$ may only contain segmentations that are equivalent to a Pareto undominated segmentation S . If some S' that is equivalent to S is not in $\mathcal{S}$, then no segmentation in $\mathcal{S}$ blocks S' so external stability is violated. So $\mathcal{S}$ must contain *all* segmentations that are equivalent to a Pareto undominated segmentation S , which we can assume to be canonical without loss of generality. To complete the necessity direction, it remains to show that the canonical segmentation S weakly blocks any non-equivalent segmentation.

Suppose for contradiction that S does not weakly block some non-equivalent segmentation S' . Because $\mathcal{S}$ is a stable set and contains all segmentations that are equivalent to S , there is a segmentation S'' that blocks S' and is equivalent to S . Consider a segment (C'', p) in

S'' that objects to S' , and the unique segment (C, p) in S in which the price is p . Because (C'', p) objects to S' and $C'' \subseteq C$, there is a positive measure of consumers in (C, p) that strictly prefer S to S' . Because S does not weakly block S' , there exists some optimal price v for C such that all consumers with value v in C strictly prefer S' to (C, p) . We claim that v is also optimal for any segment in S'' with price p , and therefore for C'' . To see this, consider all the segments $(C''_1, p), \dots, (C''_k, p)$ in S'' with price is p , so C is the union of all these coalitions, one of which is C'' . Because p is optimal for C''_j , $j = 1, \dots, k$, we have $vF^{C''_j}(v) \leq pF^{C''_j}(p)$. If price v is not optimal for some C''_j , then $vF^{C''_j}(v) < pF^{C''_j}(p)$. In this case, summing up over all j , we have $vF^C(v) < pF^C(p)$, which contradicts the optimality of price p for coalition C . So v must be optimal for C'' . Therefore, v is optimal for C'' , which means C'' contains a positive measure of consumers with value v . And because all consumers with value v in C strictly prefer S' to S , and S'' is equivalent to S' , all these consumers strictly prefer S' to (C'', p) so (C'', p) cannot object to S' , a contradiction.

To establish sufficiency, consider any canonical segmentation $S = \{(C_1, v_1), \dots, (C_n, v_n)\}$ that weakly blocks any non-equivalent segmentation. The set of segmentations that are equivalent to S satisfies internal stability because no segmentation blocks an equivalent segmentation. For external stability, we show that for any segmentation S' that is not equivalent to S , there is a segmentation S'' that is equivalent to S and blocks S' .

Consider the segment (C, p) in S that weakly objects to S' . We will construct a coalition $C'' \subseteq C$ and show that the segmentation S'' that is the same as S except that (C, p) is replaced with $(C \setminus C'', p)$ and (C'', p) , and is therefore equivalent to S , objects to S' . The construction of C'' has two steps. First, let C''_1 be a small coalition that comprises consumers with all the values that are optimal prices for C in proportions that make these values optimal prices for C''_1 . That is, $\epsilon = vF^{C''_1}(v)$ for some small ϵ and all v that are optimal prices for C . For the second step, let v' be such that a positive measure of consumers in C with value v' strictly prefer (C, p) to S' . For some $\delta > 0$, add to C''_1 a measure δ of consumers in C with value v' that strictly prefer (C, p) to S' , and remove from C''_1 the same measure δ of consumers with the highest value in C''_1 that is at most v' (a positive measure of these consumers exists because some consumers in C''_1 have value p and $p < v'$, otherwise consumers with value v' have zero surplus in S so do not strictly prefer (C, p) to S'). The resulting coalition is C'' , which, if δ is small relative to ϵ , satisfies that $\epsilon = vF^{C''}(v)$ for all prices v that are optimal for C . So if δ is small relative to ϵ , then C'' has the same set of optimal prices as C''_1 , and (C'', p) is a segment. Similarly, if ϵ and δ are small enough, then $C \setminus C''$ has the same set of optimal prices as C , so $(C \setminus C'', p)$ is a segment. By construction, (C'', p) objects to S' , so S'' , which is equivalent to S , blocks S' . ■

The canonical segmentation S in Example 7 weakly blocks any non-equivalent segmen-

tation, so the set of segmentations that are equivalent to S is a stable set. To see that S weakly blocks any non-equivalent segmentation, consider some segmentation S' that is not weakly blocked by S . Suppose without loss of generality that S' is canonical, so $S' = \{(C'_1, 1), (C'_2, 2), (C'_3, 3)\}$. Because all consumers with value 1 have zero surplus, if some consumers with value 3 in C_1 strictly preferred S to S' , then $(C_1, 1)$ would weakly object to S' . Thus, the consumers with value 3 in C_1 are in C'_1 . It is impossible for all value 2 consumers in C_2 to be in C'_1 , because then the revenue from price 2, $2 \cdot (\frac{4}{21} + \frac{11}{21})$, would be strictly higher than the revenue from price 1, which is at most $\frac{6}{21} + \frac{4}{21} + \frac{11}{21}$. Thus, some consumers with value 2 in C_2 weakly prefer S to S' . This implies that the consumers with value 3 in C_2 must be in $C'_1 \cup C'_2$, otherwise some such consumers, those in C'_3 , would strictly prefer S to S' and then $(C_2, 2)$ would weakly object to S' . Therefore C'_3 is empty (C'_3 cannot only contain consumers with value other than 3 because then price 3 cannot be optimal). Let $\delta_1 \geq 0$ be the measure of value 1 consumers in C'_2 , and let $\delta_2, \delta_3 \geq 0$ be the measures of value 2 and value 3 consumers from C_2 that are in C'_1 . For price 1 to be optimal for C'_1 , the revenue from this price, $\frac{6}{21} - \delta_1 + \delta_2 + \frac{3}{21} + \delta_3$, must be no lower than the revenue from price 3, $3 \cdot (\frac{3}{21} + \delta_3)$, which means that $-\delta_1 + \delta_2 + \delta_3 \geq 3\delta_3$. Similarly, for price 2 to be optimal for C'_2 we must have $2 \cdot (\frac{4}{21} - \delta_2 + \frac{8}{21} - \delta_3) \geq 3 \cdot (\frac{8}{21} - \delta_3)$, which means that $3\delta_3 \geq 2(\delta_2 + \delta_3)$. These two inequalities hold if and only if $\delta_1 = \delta_2 = \delta_3 = 0$. We therefore have that $C_1 = C'_1$ and $C_2 = C'_2$, so S' is equivalent to S .

H.2 Farsighted Stable Set

Our stability notion is also related to two other notions motivated by farsighted stability: the Harsanyi stable set and the Ray and Vohra farsighted stable set (henceforth RV stable set).

Both notions define a stable set as one that satisfies internal and external stability, just like the stable set of Morgenstern and Von Neumann (1953). But the notion of blocking used to define internal and external stability is “farsighted.” A segmentation blocks another segmentation if there is a sequence of segmentations that begins with the segmentation to be blocked and ends with the blocking segmentation such that each intermediate segmentation contains a coalition that prefers the blocking segmentation to the one that preceded the intermediate segmentation. These objecting coalitions allow the blocking segmentation to be “reached” starting from the original segmentation. The two notions differ in what is assumed about the segments along the sequence other than the objecting segments, with the RV stable set assuming a kind of “coalitional autonomy” similar to the “coalitional IR” that motivates our definition of stability.

Our notion of stability satisfies these two notions which, although differing in general, coincide in our setting. Moreover, although these are set notions, in our setting they are satisfied only by singleton sets. Importantly, however, these notions are not particularly useful in our setting because they are too permissive. More precisely, for each notion we have a weak and a strong version; any segmentation that does not eliminate all consumer surplus satisfies the weak versions, and any Pareto undominated segmentation satisfied the strong versions. We provide the details below.

H.2.1 Supporting Analysis for the Farsighted Stable Set

To apply the Harsanyi and RV stable sets to our setting, we need to address two technical issues. First, these notions are defined for a finite number of players. Second, they involve a definition of objection that requires a strict improvement for all members of the objecting coalition. In our setting, consumers with the lowest value in a coalition have zero surplus, so these notions become trivial (every segmentation satisfies them) if we require a strict improvement for every consumer. We define modified versions of these notions below, allowing for a continuum of players and weak improvements. Because farsighted stability considers sequences of deviations, there are two ways to allow for weak improvements. We therefore define two versions of each solution concept.

Definition 16 *A segmentation S Harsanyi blocks a segmentation S' if there is a sequence $S^0 = S', S^1, \dots, S^n = S$ of segmentations and a sequence $(C^1, p^1), \dots, (C^n, p^n)$ of segments such that for $i = 1 \dots n$, $(C^i, p^i) \in S^i$ and $CS(c, S^{i-1}) \leq CS(c, S)$ for all consumers $c \in C^i$, with a strict inequality for a positive measure of consumers $c \in C^i$ for some i . If, in addition, $CS(c, S^{i-1}) < CS(c, S)$ for a positive measure of consumers $c \in C^i$ for all $i = 1 \dots n$, we say that S strongly Harsanyi blocks S' .*

Definition 17 *A set of segmentations $\mathcal{S}$ is a (strong) Harsanyi stable set if it satisfies the following two properties:*

1. *Internal Stability: For all $S \in \mathcal{S}$, there exists no $S' \in \mathcal{S}$ that (strong) Harsanyi blocks S .*
2. *External Stability: For all $S \notin \mathcal{S}$, there exists $S' \in \mathcal{S}$ that (strong) Harsanyi blocks S .*

Definition 18 *A segmentation S RV blocks a segmentation S' if there is a sequence $S^0 = S', S^1, \dots, S^n = S$ of segmentations and a sequence $(C^1, p^1), \dots, (C^n, p^n)$ of segments such that for $i = 1 \dots n$, $(C^i, p^i) \in S^i$ and $(C, p) \in S^i$ whenever $(C, p) \in S^{i-1}$ and $C \cap C^i = \emptyset$, and $CS(c, S^{i-1}) \leq CS(c, S)$ for all consumers $c \in C^i$, with a strict inequality for a positive*

measure of consumers $c \in C^i$ for some i . If, in addition, $CS(c, S^{i-1}) < CS(c, S)$ for a positive measure of consumers $c \in C^i$ for all $i = 1 \dots n$, we say that S strongly RV blocks S' .

Definition 19 A set of segmentations $\mathcal{S}$ is a (strong) RV stable set if it satisfies the following two properties:

1. *Internal Stability:* For all $S \in \mathcal{S}$, there exists no $S' \in \mathcal{S}$ that (strong) RV blocks S .
2. *External Stability:* For all $S \notin \mathcal{S}$, there exists $S' \in \mathcal{S}$ that (strong) RV blocks S .

For the following characterization of Harsanyi and RV stable sets we denote by $ACS(S)$ the average consumer surplus in segmentation S .

Proposition 7 The following are equivalent for any set of segmentations $\mathcal{S}$:

- $\mathcal{S}$ is a Harsanyi stable set
- $\mathcal{S}$ is a RV stable set
- $\mathcal{S} = \{S\}$ for some S with $ACS(S) > 0$.

The proof of Proposition 7 uses the following lemma.

Lemma 4 For any two segmentations S and S' , the following are equivalent:

- S Harsanyi blocks S' .
- S RV blocks S' .
- $ACS(S) > 0$.

Proof. If $ACS(S) = 0$, then $CS(c, S) = 0$ for all consumers. Therefore, S cannot Harsanyi block or RV block any segmentation.

Suppose that $ACS(S) > 0$. We show that S RV blocks any segmentation S' , which also implies that S Harsanyi blocks S' . We do so by constructing a sequence of segmentations in several steps that gradually transform S' to an elementary segmentation in which each segment includes consumers with a single value. We then proceed from the elementary segmentation to S .

In step 0 we set $S^0 = S'$. In each following step $i > 0$, we take the segmentation S^{i-1} and a segment (C, p) in S^{i-1} that contains consumers of at least two types. For each value v_j , we let $C_{v_j}^i$ be the set of all consumers with value v_j in C . S^i is constructed from S^{i-1}

by replacing (C, p) with the segments $(C_{v_j}^i, v_j)$ for all j such that $f^C(v_j) > 0$. Let $C^i = C_p^i$. The first phase ends with a segmentation in which every segment contains consumers of only a single type, so the surplus of all consumers is zero. The second phase has one step per segment in S . In particular, for each (C, p) in S , given S^{i-1} , we remove the consumers in C from the segments in S^{i-1} with single values and the new segment (C, p) to construct S^i with $C^i = C$. The second phase ends with $S^n = S$.

To see that S RV blocks S' , notice that in each step $i = 1, \dots, n$, consumers in C^i have zero surplus in S^{i-1} . Therefore, they weakly prefer S to S^{i-1} . Additionally, because $ACS(S) > 0$, there is a segment (C, p) in S in which a positive measure of consumers obtain positive surplus. As a result, a positive measure of consumers strictly prefer $S = S^n$ to S^{n-1} .

■

Proof of Proposition 7. Suppose that $\mathcal{S} = \{S\}$ for some S with $ACS(S) > 0$. Then, by Lemma 4, S RV blocks and Harsanyi blocks any $S' \neq S$, so $\mathcal{S}$ is a RV stable set and a Harsanyi stable set.

Consider any Harsanyi (respectively RV) stable set $\mathcal{S}$. The set $\mathcal{S}$ must contain at least one segmentation S with $ACS(S) > 0$, otherwise a segmentation $S' \notin \mathcal{S}$ is not Harsanyi (RV) blocked by any segmentation in $\mathcal{S}$ by Lemma 4. If the set contains more than one segmentation, then, by Lemma 4, the segmentation S that satisfies $ACS(S) > 0$ Harsanyi (RV) blocks the other segmentations in the set. Therefore, $\mathcal{S}$ contains a single segmentation S , and $ACS(S) > 0$. ■

Proposition 8 *The following are equivalent for any set of segmentations $\mathcal{S}$:*

- $\mathcal{S}$ is a strong Harsanyi stable set
- $\mathcal{S}$ is a strong RV stable set
- $\mathcal{S}$ is the set of all segmentations that are equivalent to some segmentation S that is Pareto undominated.

The proof uses the following lemma. The lemma uses a weak notion of equivalence of segmentations. Namely, we say that two segmentations S and S' are surplus-equivalent if almost all consumers have the same surplus in the two segmentations, that is, for almost all $c \in [0, 1]$, $CS(c, S) = CS(c, S')$. Any two equivalent segmentations are surplus-equivalent.

Lemma 5 *For any two segmentations S and S' , the following are equivalent:*

- Some surplus-equivalent segmentation to S strong Harsanyi blocks S' .
- Some surplus-equivalent segmentation to S strong RV blocks S' .

- *There exist a positive measure of consumers c such that $CS(c, S) > CS(c, S')$.*

Proof. If some segmentation S'' that is surplus-equivalent to S strong Harsanyi (RV) blocks S' , then, by definition, a positive measure of consumers strictly prefer S'' , and therefore S , to S' .

Suppose that a positive measure of consumers strictly prefer S to S' . We show that some segmentation S'' that is surplus-equivalent to S strong RV blocks segmentation S' , which also implies that S'' strong Harsanyi blocks S' .

We do so by constructing a sequence of segmentations in two phases. The first phase consists of two steps. In the first step, consider some segment (C, p) in S that contains a positive measure of consumers that strictly prefer S to S' . Let coalition C^1 contain a positive measure of consumers with value p from C , a positive measure of (but not all the) consumers from C that strictly prefer S to S' , and, for *every* segment in S' , a positive measure of consumers with the lowest value in that segment, where the proportions of consumers in C^1 are such that (C^1, p) is a segment. Consider a segmentation S^1 that consists of (C^1, p) and, for each consumer value, a segment that contains only the consumers in $[0, 1] \setminus C^1$ with that value, so their surplus is zero. In the second step, replace (C^1, p) with (C, p) and, for each consumer value, put the consumers in $C^1 \setminus C$ with that value in a separate segment (all other segments remain intact). Denote the resulting segmentation by S^0 .

The second phase consists of (potentially) several steps. In each step $i > 0$, take segmentation S^{i-1} and, for some segment (C', p') in S that is not already in S^{i-1} and contains a positive measure of consumers with positive surplus, let $C^i = C'$. S^i is constructed from S^{i-1} by taking all segments (C'', p'') that contain a positive measure of consumers from C' (so C'' contains only consumers with value p'') and replacing them with $(C'' \setminus C', p'')$, and finally adding segment (C', p') to S^i . This process ends with a final segmentation S^n that may differ from S but is surplus-equivalent to it because for any segment in S that is not in S^n , all consumers in that segment obtain zero surplus in both segmentations. So, for the remainder of the proof, suppose without loss of generality that $S = S^n$.

To see that S RV blocks S' , notice that in the first step of the first phase, coalition C^1 contains some consumers that strictly prefer S to S' , and all other consumers in C^1 weakly prefer S to S' because they have surplus zero in S' . In the second step of the first phase, by definition, some consumers in C strictly prefer S to S^1 and all the consumers in $C \setminus C^1$ because they have zero surplus in S^1 . Similarly, in each step i of the second phase, consumers in C^i have surplus zero in S^{i-1} , and some consumers in C^i strictly prefer S to S^{i-1} because they have a positive surplus in S . ■

Proof of Proposition 8. We first show for any Pareto undominated S , the set of segmentations that are equivalent to S is the same as the set of segmentations that are

surplus-equivalent to S . For this, we show that a segmentation S' is equivalent to Pareto undominated S if and only if it is surplus-equivalent to S . If S' is equivalent to it, then almost all consumers have the same surplus in the two segmentations, and therefore they are surplus-equivalent. Suppose S and S' are surplus equivalent. Because S is Pareto undominated, so is S' . By Lemma 3, both segmentations must be efficient. Let $\{(C_1, v_1), \dots, (C_n, v_n)\}$ and $\{(C'_1, v_1), \dots, (C'_n, v_n)\}$ be the canonical representations of S and S' , respectively. Because the two segmentations are efficient, all consumers in value v_1 are in both C_1 and C'_1 . If some consumer with value higher than v_1 is in C_1 but not C'_1 , then the consumer's surplus is different across the two segmentations, violating surplus-equivalence. A similar argument applies to consumers in C'_1 but not C_1 . So we must have $C_1 = C'_1$. An inductive argument implies that the two segmentations are equivalent.

Suppose that $\mathcal{S}$ is the set of all segmentations that are equivalent, and hence surplus-equivalent, to some Pareto undominated segmentation S . Because S is Pareto undominated, for any segmentation that is not in $\mathcal{S}$, and hence is not surplus-equivalent to S , there are some consumers that strictly prefer S to S' . By Lemma 5, S strong RV blocks and strong Harsanyi blocks any such S' , so external stability is satisfied. Any two segmentations in $\mathcal{S}$ are surplus-equivalent and therefore by Lemma 5, neither strong RV blocks nor strong Harsanyi blocks the other, and therefore internal stability is satisfied. So $\mathcal{S}$ is a strong RV stable set and a strong Harsanyi stable set.

Consider a strong Harsanyi (RV) stable set $\mathcal{S}$. If the set contains two segmentations S and S' in $\mathcal{S}$ that are not surplus-equivalent, then there is a positive measure of consumers that either prefer S to S' or S' to S . Then, by Lemma 5, one of the two segmentations strong Harsanyi (RV) blocks the other one, violating internal stability. Therefore, $\mathcal{S}$ contains only surplus-equivalent segmentations. Further, if S and S' are surplus-equivalent and one of them is in $\mathcal{S}$, then the other one must be too, because otherwise again by Lemma 5 the segmentation that is not in $\mathcal{S}$ is not Harsanyi (RV) blocked by any segmentation in $\mathcal{S}$, violating external stability. A segmentation S in $\mathcal{S}$ cannot be Pareto dominated by any segmentation S' not in $\mathcal{S}$ because otherwise, by Lemma 5, S would not strong Harsanyi (RV) block S' , violating external stability. So $\mathcal{S}$ must be the set of all surplus-equivalent segmentations, and therefore equivalent segmentations, to the Pareto undominated segmentation S . ■

The stable set notions we study in this section allow for weak inequalities along a sequence of segmentations. If we require strict improvement in every step, then no segmentation can block any other segmentation because in any segmentation some consumers get zero surplus. So in that case, the unique RV stable set (and also Harsanyi and also maximal RV stable set) is the set of all segmentations.

Ray and Vohra (2019) define a notion of *maximality* of a stable set and show that any single-payoff RV stable set is also a maximal RV stable set. Because both Proposition 7 and Proposition 8 characterize stable sets as ones that contain a single segmentation (or surplus-equivalent ones), those stable sets are also maximal.³¹ Roughly speaking, maximality requires that in a chain of segmentations defined in Definition 18 that ends in S , at each step the move specified by the chain is “optimal” in the sense that no coalition C has another move that would lead to another segmentation in the stable set that the coalition C prefers to S . If the stable set is a singleton, then all chains necessarily end in the same segmentation, and therefore maximality is trivially satisfied.

³¹Theorem 1 together with Remark 1 in Ray and Vohra (2019) show that in their setting with a finite number of players and objections that are defined to require strict improvements, any singleton RV stable set is also a maximal RV stable set. A similar argument shows that this is also the case in our setting.